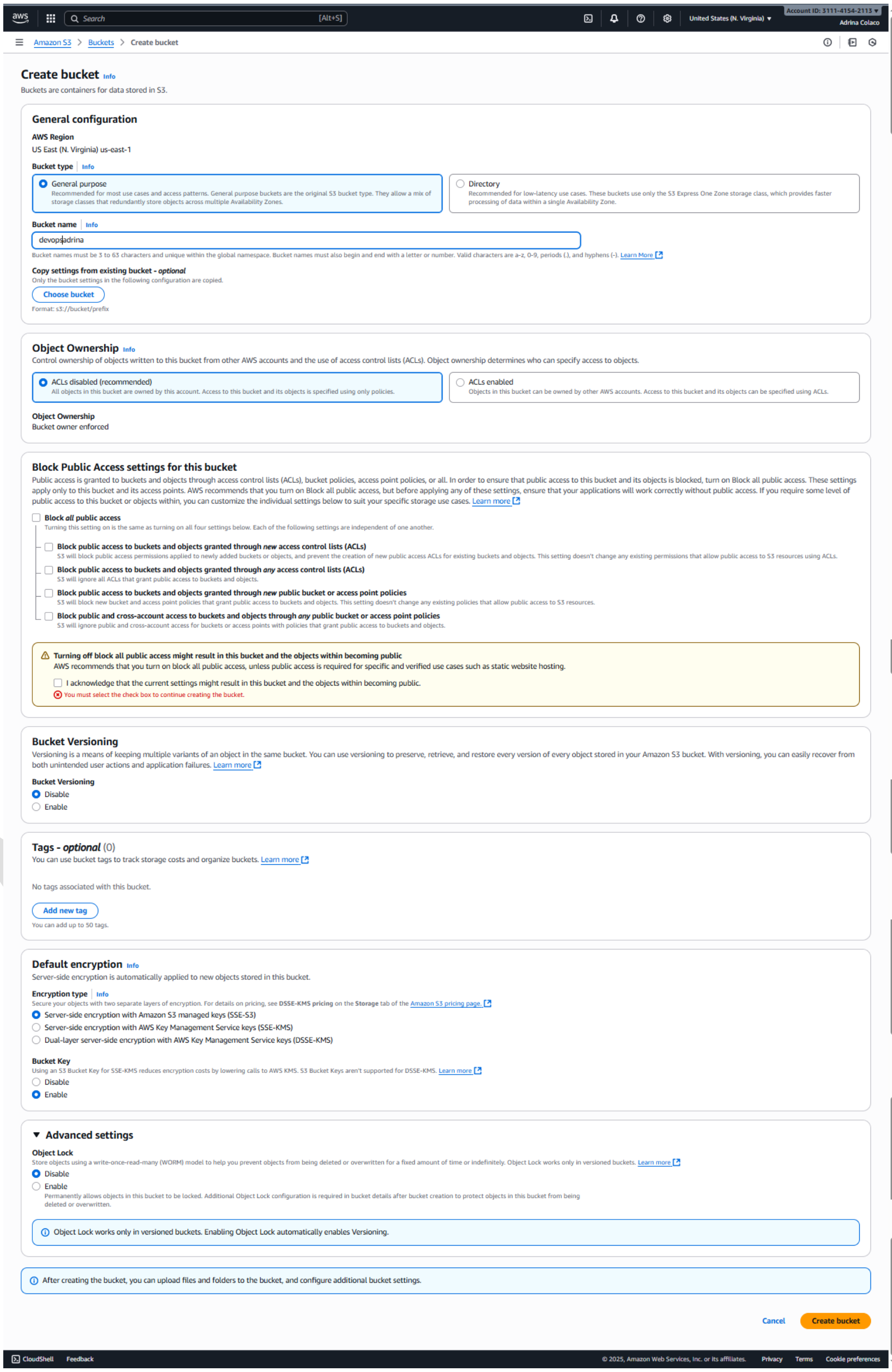
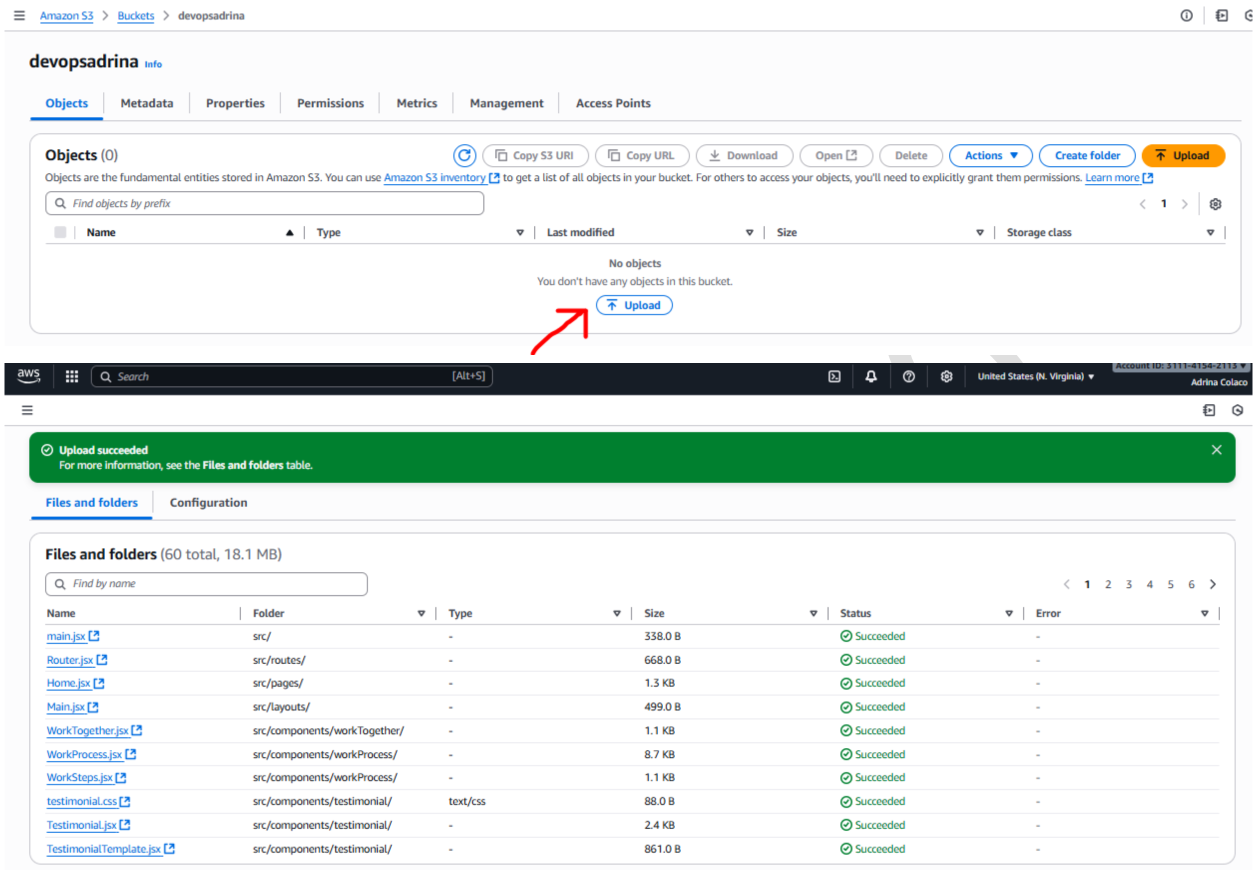
AWS- S3

Assignment 14 – Static Website Setup

1. Create a bucket with unique name.

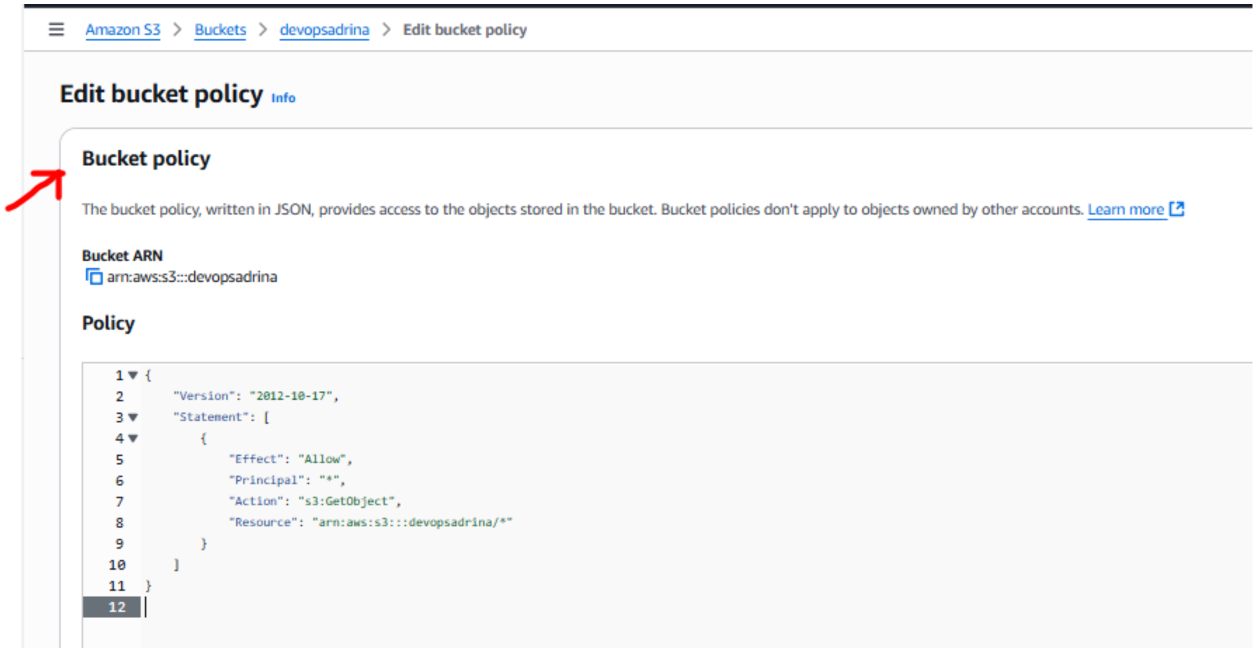


2. Upload the website files to S3 bucket.

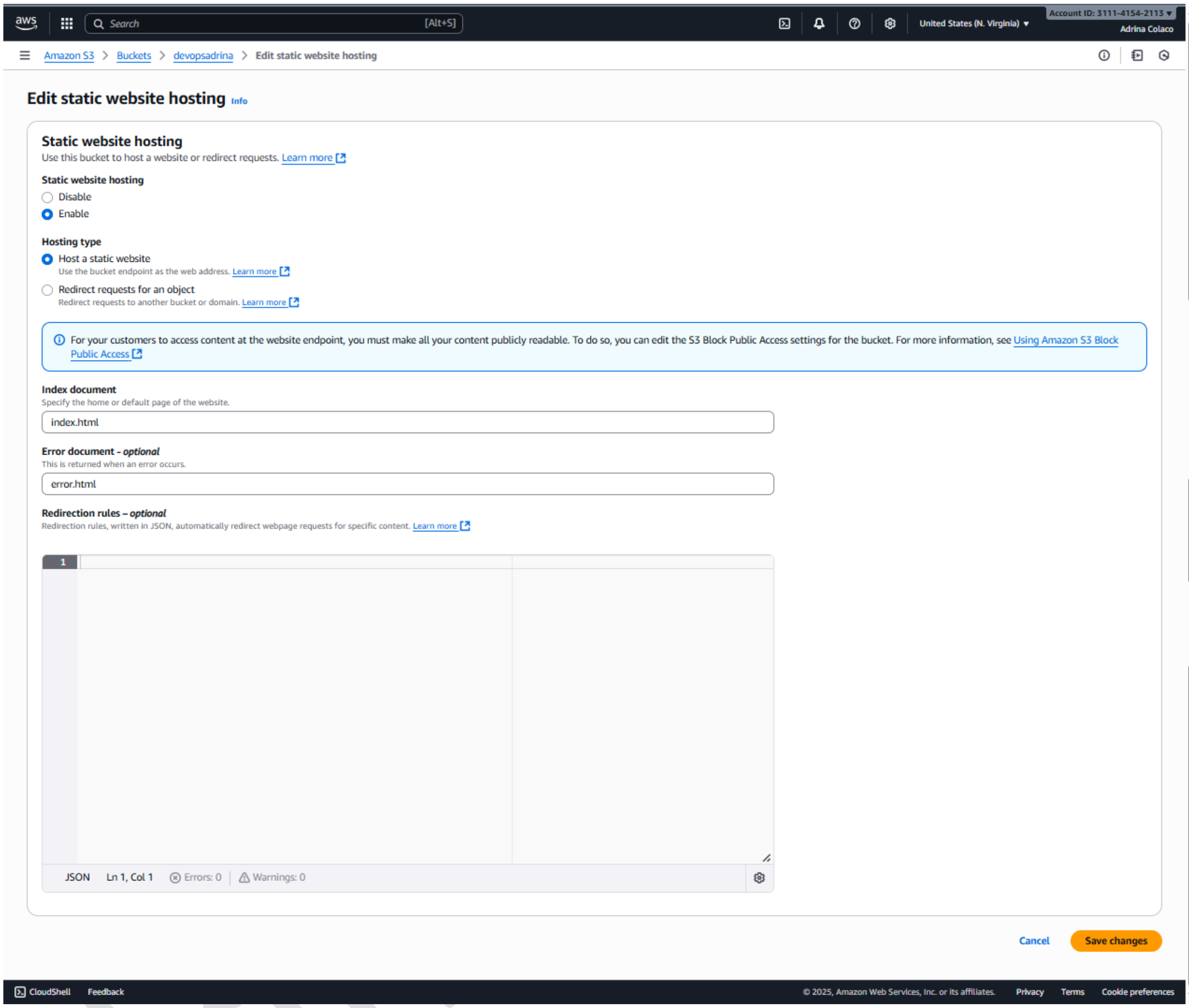


3. Change bucket policy in Permissions

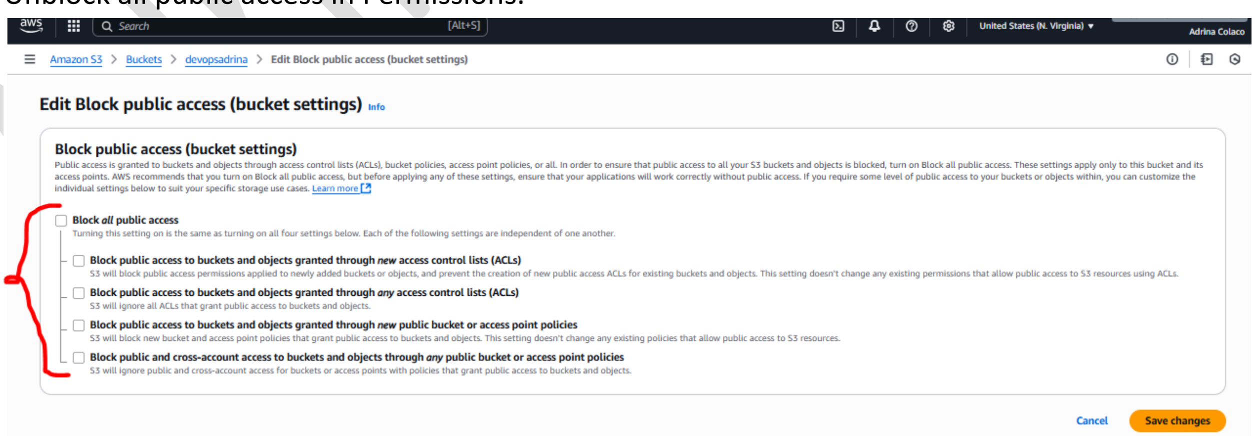
```json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::devopsadrina/*"
    }
  ]
}
```



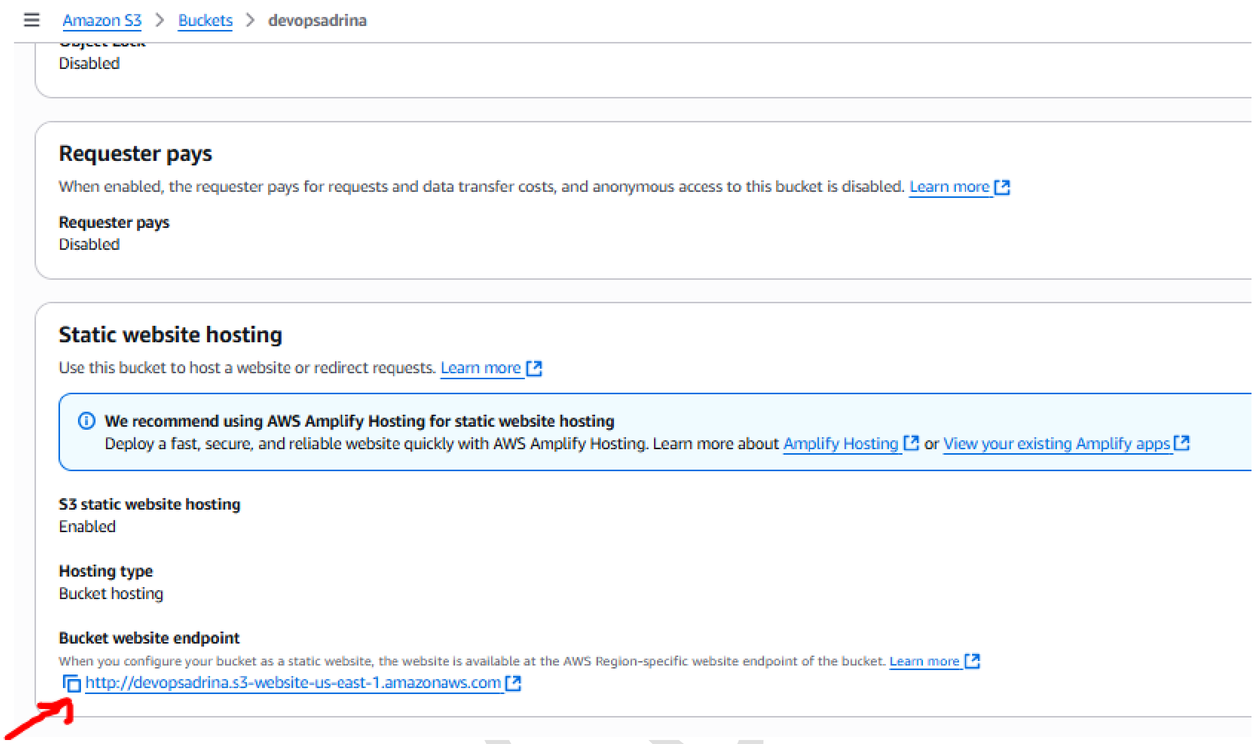
4. Change the Properties to enable static website hosting.



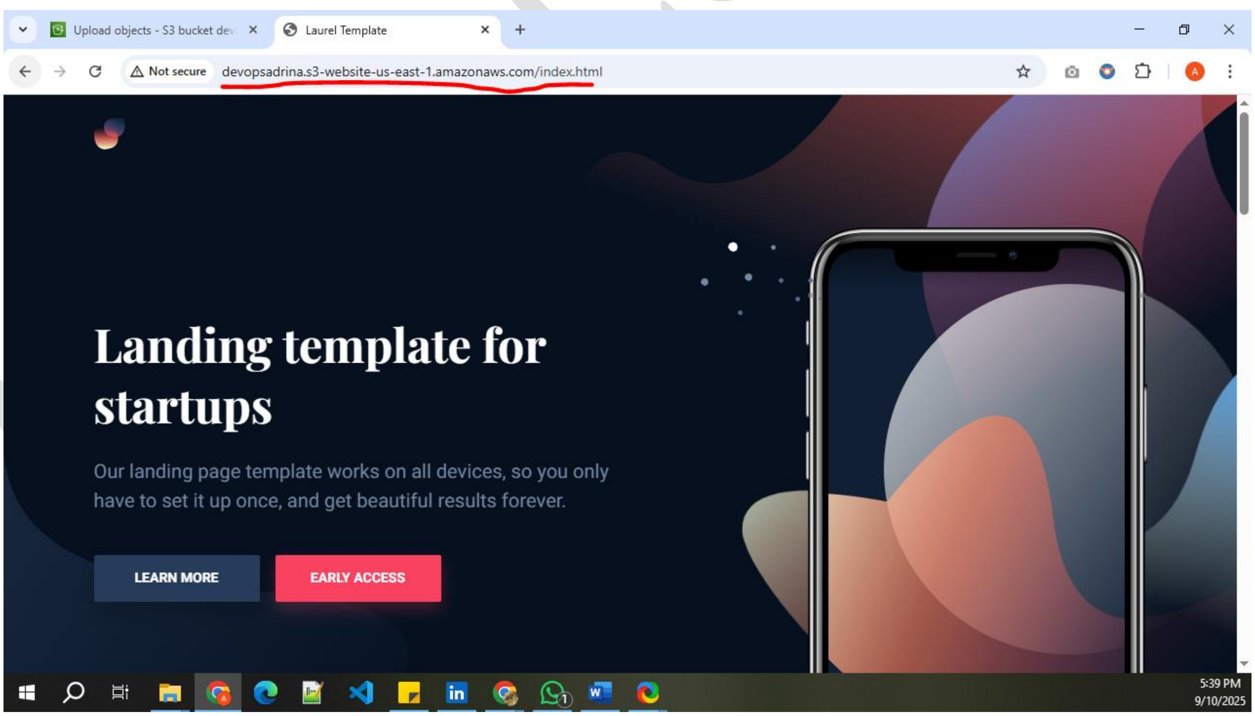
5. Unblock all public access in Permissions.



6. Verify that the website is working by running the Bucket website endpoint URL into a browser.



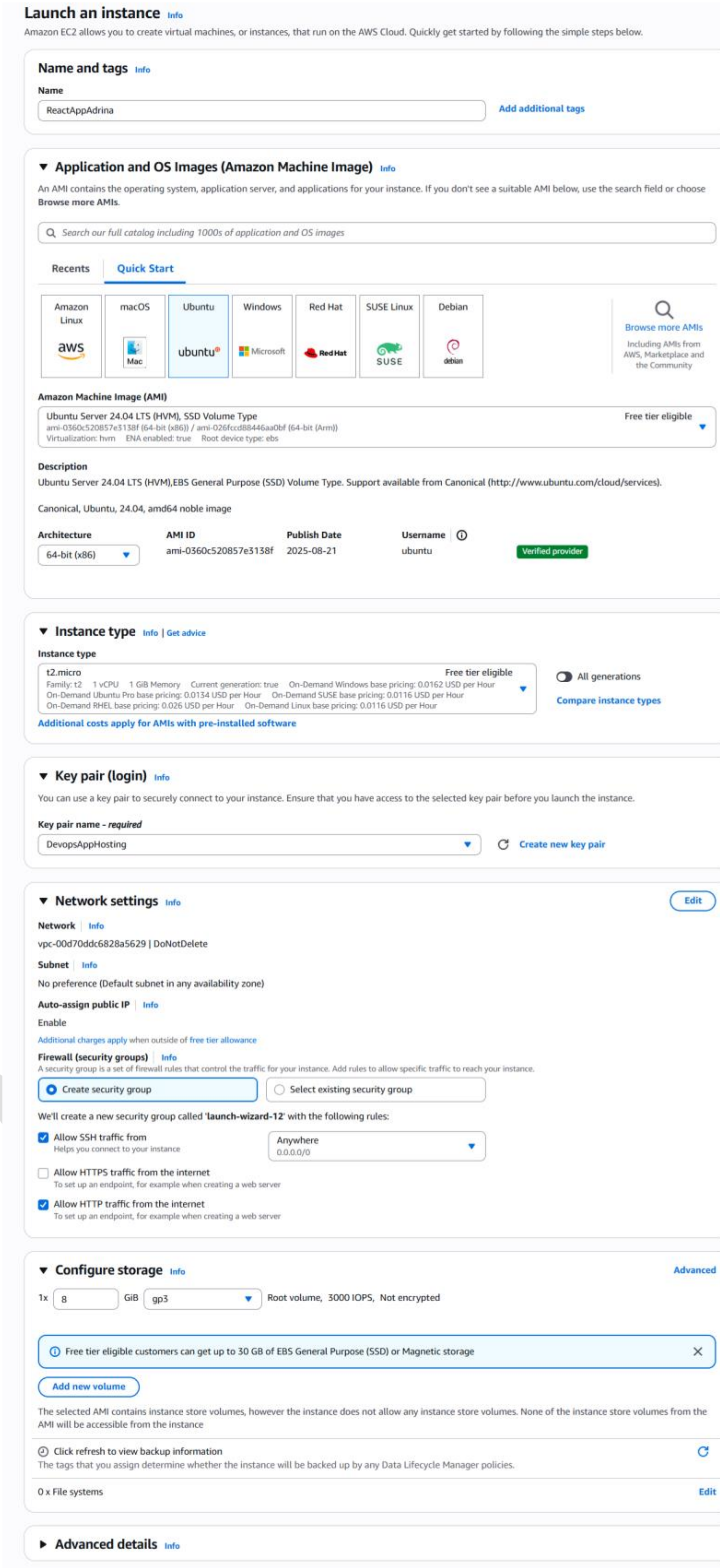
Amazon S3 > Buckets > devopsadrina
Disabled
Requester pays
When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. Learn more
Requester pays
Disabled
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps
S3 static website hosting
Enabled
Hosting type
Bucket hosting
Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more
http://devopsadrina.s3-website-us-east-1.amazonaws.com

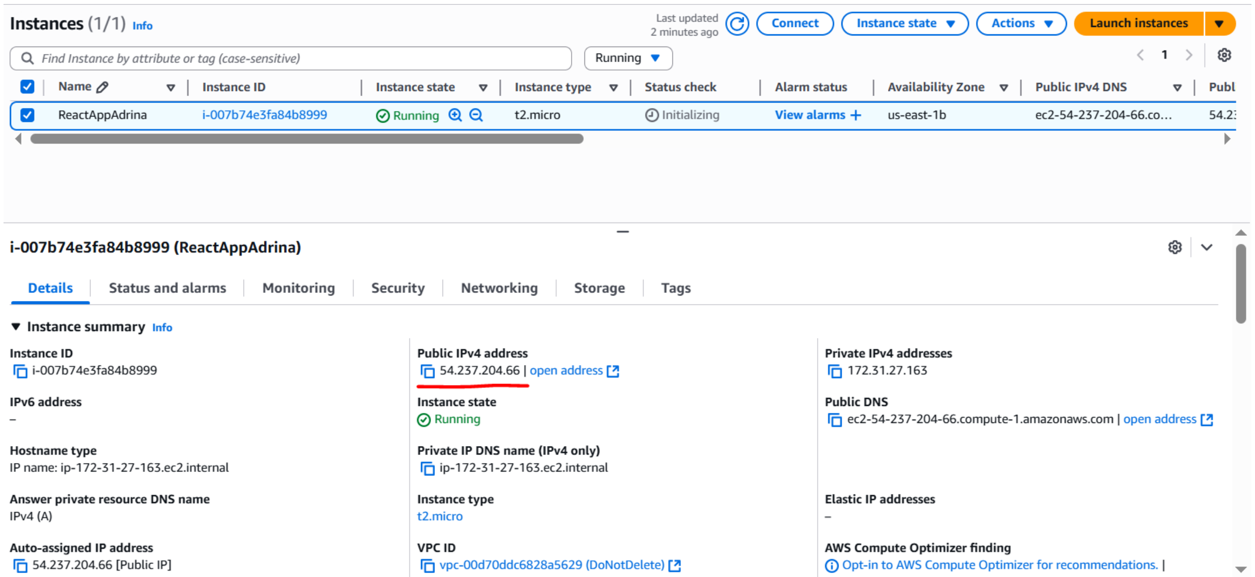


Upload objects - S3 bucket dev
Laurel Template
Not secure
devopsadrina.s3-website-us-east-1.amazonaws.com/index.html
Landing template for startups
Our landing page template works on all devices, so you only have to set it up once, and get beautiful results forever.
LEARN MORE
EARLY ACCESS
5:39 PM
9/10/2025

Assignment 15- Deploy React Application On Virtual Machine

1. Launch an EC2 instance





2. Login EC2

3. Update packages, Install Nodejs and npm
   sudo apt update
   sudo apt install -y nodejs npm

```
ubuntu@ip-172-31-27-163:~$ node -v
v18.19.1
ubuntu@ip-172-31-27-163:~$ npm -v
9.2.0
ubuntu@ip-172-31-27-163:~$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

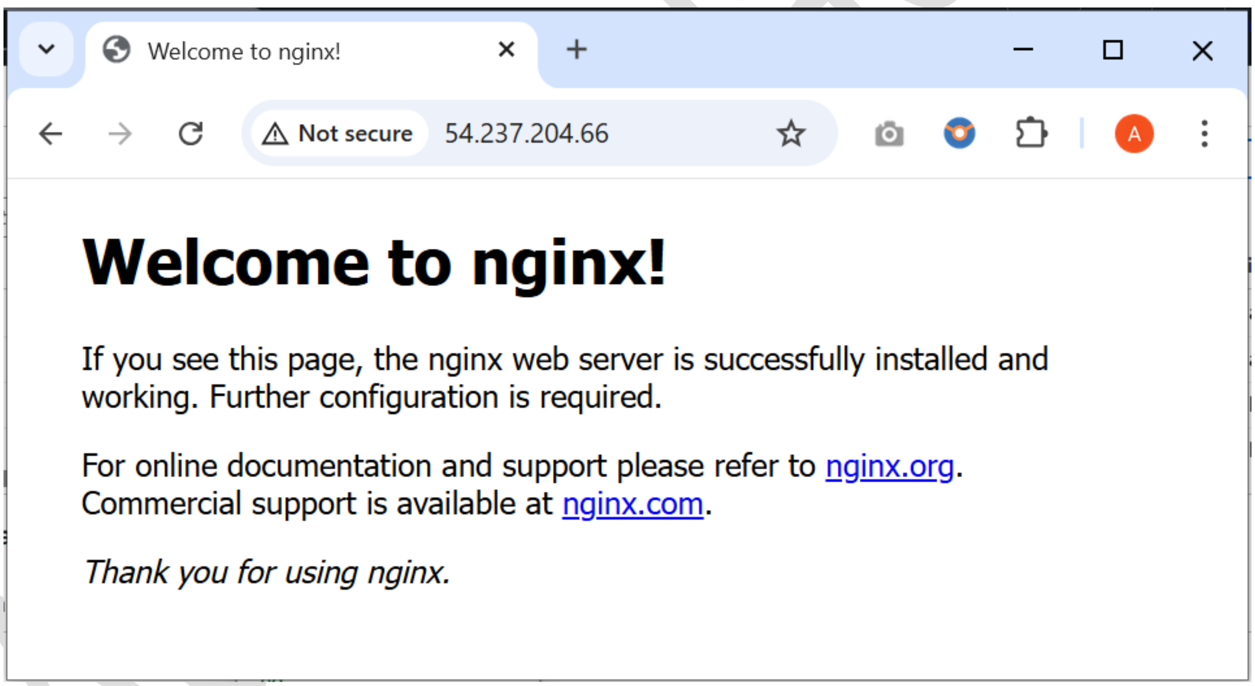
4. Install and Enable NGINX

```
ubuntu@ip-172-31-27-163:~$ sudo systemctl start nginx
ubuntu@ip-172-31-27-163:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
ubuntu@ip-172-31-27-163:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-09-18 03:58:06 UTC; 56s ago
       Docs: man:nginx(8)
   Main PID: 8050 (nginx)
      Tasks: 2 (limit: 1121)
     Memory: 1.7M (peak: 3.7M)
        CPU: 15ms
     CGroup: /system.slice/nginx.service
             ├─8050 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─8053 "nginx: worker process"

Sep 18 03:58:06 ip-172-31-27-163 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 18 03:58:06 ip-172-31-27-163 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-27-163:~$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66 PrivateIPs: 172.31.27.163

5. Check if you can access NGINX from the Public IP of the instance



6. Clone the React Application from GitHub and modify App.js.

```
ubuntu@ip-172-31-27-163:~$ git clone https://github.com/pravinmishraaws/my-react-app.git
Cloning into 'my-react-app'...
remote: Enumerating objects: 69, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (66/66), done.
remote: Total 69 (delta 28), reused 26 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (69/69), 188.95 KiB | 13.50 MiB/s, done.
Resolving deltas: 100% (28/28), done.
ubuntu@ip-172-31-27-163:~$ cd my-react-app/
ubuntu@ip-172-31-27-163:~/my-react-app$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

```
  GNU nano 7.2                                                                              App.js
import React from "react";

function App() {
  return (
    <div style={{ textAlign: "center", paddingTop: "50px" }}>
      <h1>Welcome to My React App</h1>
      <p>This app is running on Nginx!</p>

      <h2>Deployed by: <strong>Adrina Colaco</strong></h2>
      <p>Date: <strong>18/9/2025</strong></p>

      <hr style={{ margin: "20px 0" }} />

      <p>
        P.S. This post is part of the FREE DevOps for Beginners Cohort run by
        <strong> Pravin Mishra</strong>.
      </p>
      <p>
        You can start your DevOps journey for free from his{" "}
        <a
          href="https://www.youtube.com/playlist?list=PLVOdqXbCs7bX88JeUZmK4fKTq2hJ5VS89"
          target="_blank"
          rel="noopener noreferrer"
        >
          YouTube Playlist
        </a>.
      </p>
      <p>
                                                                                   [ Read 43 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location      M-
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

7. Install Dependencies and Build the React App

   npm install

```
npm run build
```

```
ubuntu@ip-172-31-27-163:~/my-react-app/src$ npm run build

> my-react-app@0.1.0 build
> react-scripts build

Creating an optimized production build...
Browserslist: browsers data (caniuse-lite) is 7 months old. Please run:
  npx update-browserslist-db@latest
  Why you should do it regularly: https://github.com/browserslist/update-db#readme
Compiled successfully.

File sizes after gzip:

  59.21 kB  build/static/js/main.6ee106ef.js
  1.78 kB   build/static/js/453.ad6eb26e.chunk.js
  264 B     build/static/css/main.e6c13ad2.css

The project was built assuming it is hosted at /.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.
You may serve it with a static server:

  npm install -g serve
  serve -s build

Find out more about deployment here:

  https://cra.link/deployment
```

8. Deploy Build Files to Nginx Web Directory

```
ubuntu@ip-172-31-27-163:~/my-react-app/src$ sudo rm -rf /var/www/html/*
```

```
ubuntu@ip-172-31-27-163:~/my-react-app/src$ cd ..
ubuntu@ip-172-31-27-163:~/my-react-app$ ls
README.md  build  node_modules  package-lock.json  package.json  public  src
ubuntu@ip-172-31-27-163:~/my-react-app$ sudo cp -r build/* /var/www/html/
ubuntu@ip-172-31-27-163:~/my-react-app$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

```
ubuntu@ip-172-31-27-163:~/my-react-app$ sudo chown -R www-data:www-data /var/www/html
sudo chmod -R 755 /var/www/html
ubuntu@ip-172-31-27-163:~/my-react-app$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

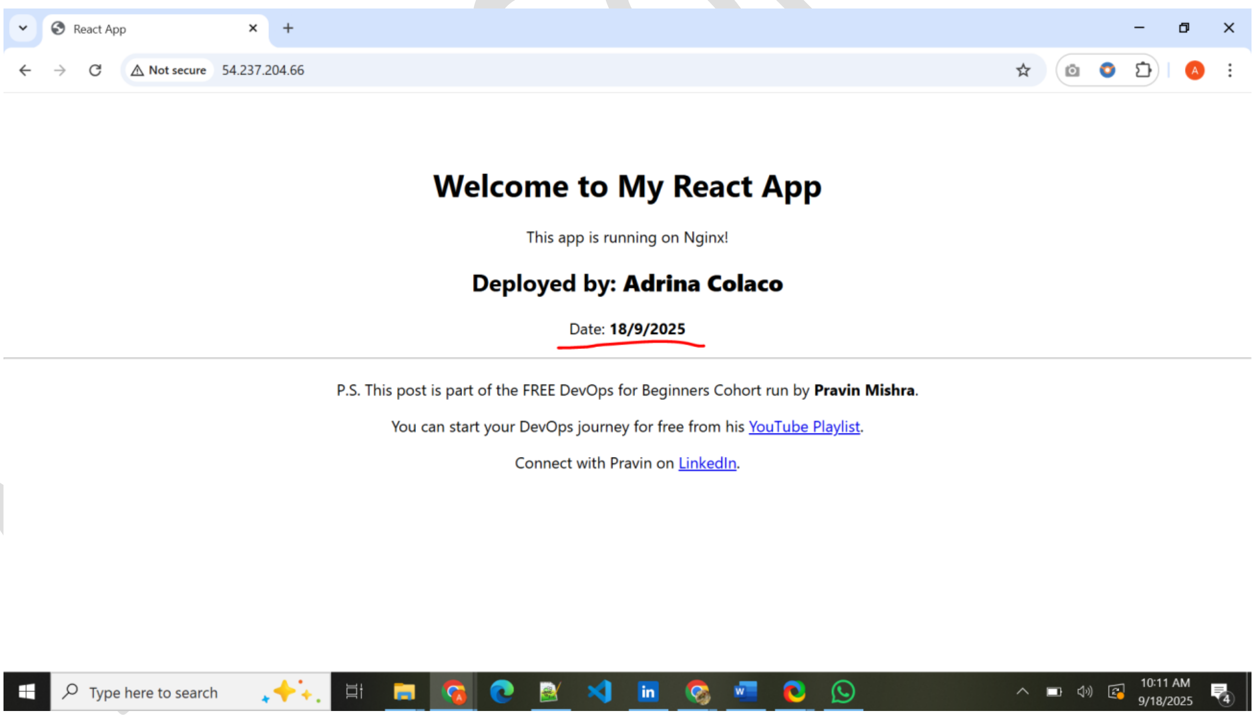
9. Configure Nginx for React

```
ubuntu@ip-172-31-27-163:~/my-react-app$ echo 'server {
    listen 80;
    server_name _;
    root /var/www/html;
    index index.html;

    location / {
        try_files $uri /index.html;
    }

    error_page 404 /index.html;
}' | sudo tee /etc/nginx/sites-available/default > /dev/null
ubuntu@ip-172-31-27-163:~/my-react-app$ sudo systemctl restart nginx
ubuntu@ip-172-31-27-163:~/my-react-app$
```

i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

10. Find Your Public IP and Access the Application



11. Verify the Deployment using CURL command

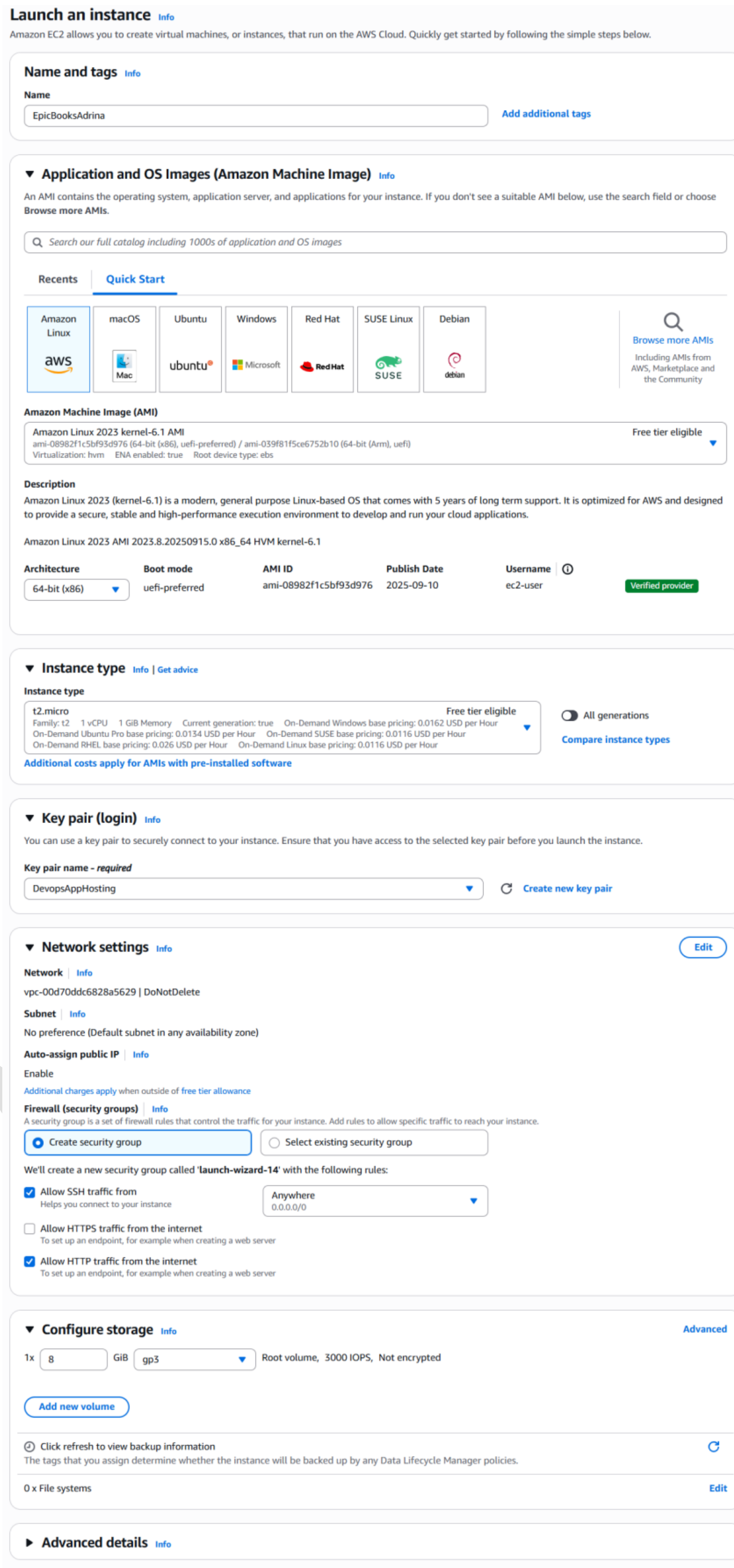
```
ubuntu@ip-172-31-27-163:~/my-react-app$ curl 54.237.204.66
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta name="theme-col
or" content="#000000"/><meta name="description" content="Web site created using create-react-app"/><link rel="apple-touch-icon" href="/logo192.png"/><link rel="manifest" href="/manifest.j
son"/><title>React App</title><script defer="defer" src="/static/js/main.6ee106ef.js"></script><link href="/static/css/main.e6c13ad2.css" rel="stylesheet"></head><body><noscript>You need
to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>ubuntu@ip-172-31-27-163:~/my-react-app$
```

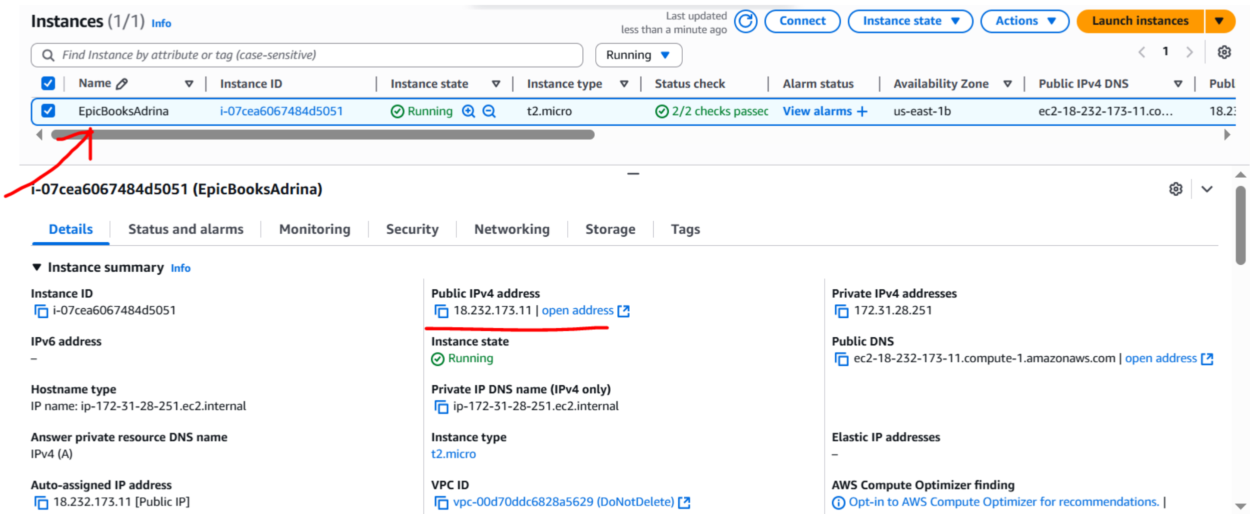
i-007b74e3fa84b8999 (ReactAppAdrina)

PublicIPs: 54.237.204.66   PrivateIPs: 172.31.27.163

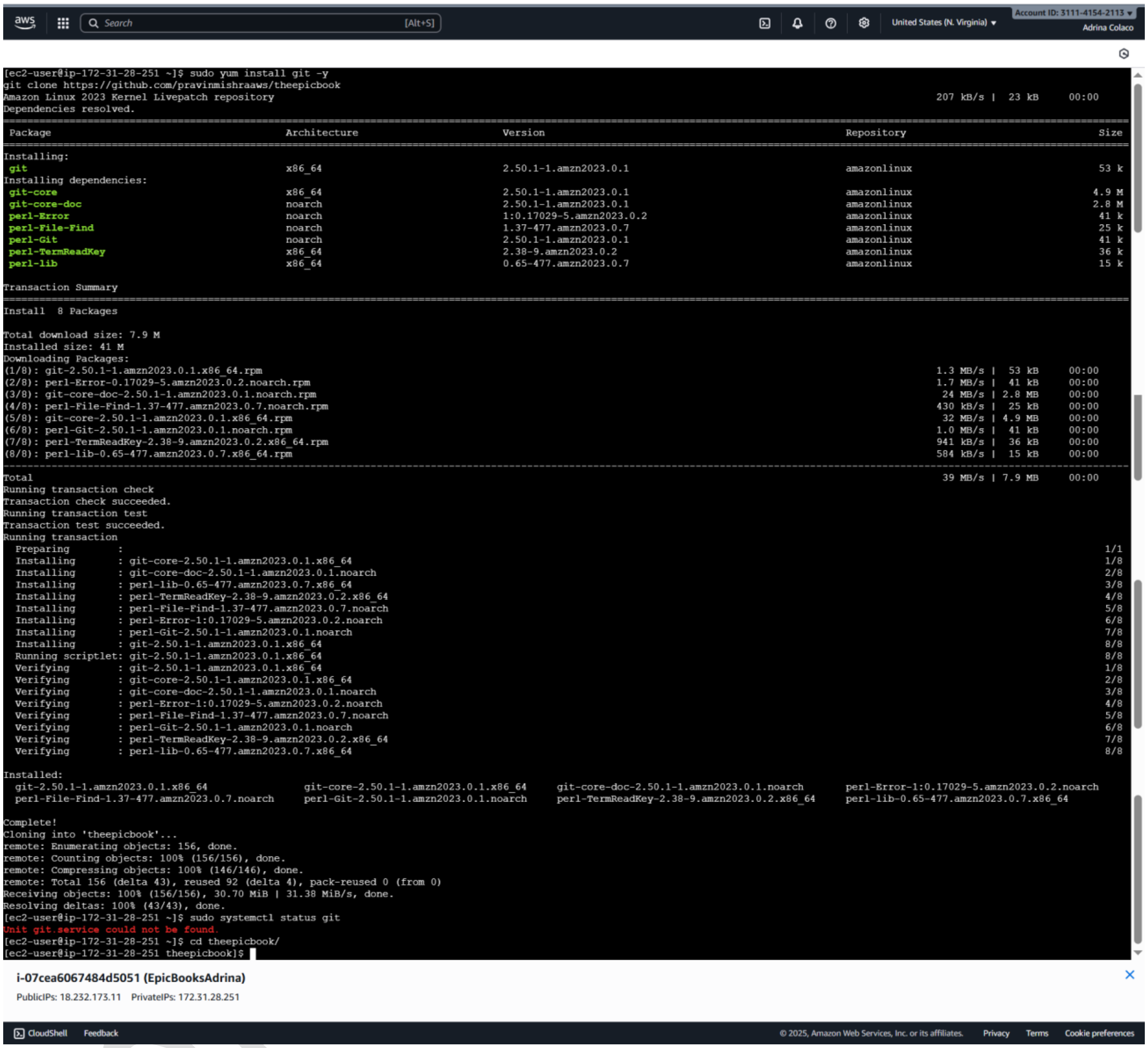
Assignment 16- Deploy EpicBooks (monolith) Application

1. Launch Ubuntu in EC2


Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
EpicBooksAdrina
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
Free tier eligible
ami-08982f1c5bf93d976 (64-bit (x86), uefi-preferred) / ami-039f81f5ce6752b10 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.8.20250915.0 x86_64 HVM kernel-6.1
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-08982f1c5bf93d976
Publish Date
2025-09-10
Username
ec2-user
Verified provider
Instance type Info | Get advice
Instance type
t2.micro
Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
DevopsAppHosting
Create new key pair
Network settings Info
Edit
Network Info
vpc-00d70ddc6828a5629 | DoNotDelete
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Additional charges apply when outside of free tier allowance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-14' with the following rules:
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
0 x File systems
Edit
Advanced details Info



2. Clone the Repository
   Install GIT
   sudo yum install git -y
   git clone https://github.com/pravinmishraaws/theepicbook



3. Install Node.js and NPM
   curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.5/install.sh | bash
   source ~/.nvm/nvm.sh
   nvm install v17
   node -v

```
[ec2-user@ip-172-31-28-251 ~]$ curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.5/install.sh | bash
source ~/.nvm/nvm.sh
nvm install v17
node -v
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 15916  100 15916    0     0   534k      0 --:--:-- --:--:-- --:--:--  555k
=> Downloading nvm from git to '/home/ec2-user/.nvm'
=> Cloning into '/home/ec2-user/.nvm'...
remote: Enumerating objects: 383, done.
remote: Counting objects: 100% (383/383), done.
remote: Compressing objects: 100% (326/326), done.
remote: Total 383 (delta 43), reused 179 (delta 29), pack-reused 0 (from 0)
Receiving objects: 100% (383/383), 391.78 KiB | 19.59 MiB/s, done.
Resolving deltas: 100% (43/43), done.
* (HEAD detached at FETCH_HEAD)
  master
=> Compressing and cleaning up git repository

=> Appending nvm source string to /home/ec2-user/.bashrc
=> Appending bash_completion source string to /home/ec2-user/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"  # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"  # This loads nvm bash_completion
Downloading and installing node v17.9.1...
Downloading https://nodejs.org/dist/v17.9.1/node-v17.9.1-linux-x64.tar.xz...
######################################################################################################################## 100.0%
Computing checksum with sha256sum
Checksums matched!
Now using node v17.9.1 (npm v8.11.0)
Creating default alias: default -> v17 (-> v17.9.1)
v17.9.1
[ec2-user@ip-172-31-28-251 ~]$
```

4. Install Project dependencies
   npm install

```
[ec2-user@ip-172-31-28-251 ~]$ npm install
npm ERR! code ENOENT
npm ERR! syscall open
npm ERR! path /home/ec2-user/package.json
npm ERR! errno -2
npm ERR! enoent ENOENT: no such file or directory, open '/home/ec2-user/package.json'
npm ERR! enoent This is related to npm not being able to find a file.
npm ERR! enoent

npm ERR! A complete log of this run can be found in:
npm ERR!     /home/ec2-user/.npm/_logs/2025-09-19T15_21_28_908Z-debug-0.log
[ec2-user@ip-172-31-28-251 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-28-251 ~]$ ls
package-lock.json  theepicbook
[ec2-user@ip-172-31-28-251 ~]$ cd theepicbook/
[ec2-user@ip-172-31-28-251 theepicbook]$ ls
'Installation & Configuration Guide.md'   config  models             package.json   routes     views
 README.md                                db      package-lock.json  public         server.js
[ec2-user@ip-172-31-28-251 theepicbook]$ npm install
npm WARN deprecated ini@1.3.5: Please update to ini >=1.3.6 to avoid a prototype pollution issue
npm WARN deprecated debug@4.1.1: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in
 a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)
npm WARN deprecated debug@3.2.6: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in
 a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)
npm WARN deprecated debug@4.1.1: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in
 a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)

added 357 packages, and audited 358 packages in 15s

30 packages are looking for funding
  run `npm fund` for details

37 vulnerabilities (5 low, 10 moderate, 18 high, 4 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 8.11.0 -> 11.6.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.0
npm notice Run npm install -g npm@11.6.0 to update!
npm notice
[ec2-user@ip-172-31-28-251 theepicbook]$
```

5. Create MySQL in AWS RDS. Install MySQL server

epicbooks

Modify | Actions

Related

Filter by databases

DB identifier	Status	Role	Engine	Region ...	Size	Recom...	CPU	Curren...
epicbooks	Available	Regional c...	Aurora My...	us-east-1	1 instance		-	-
epicbooks-instance-1	Available	Writer ins...	Aurora My...	us-east-1b	db.r7g.large		-	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations - *new*

Endpoints (2)

Actions | Create custom endpoint

Find resources

Endpoint name	Status	Type	Port
epicbooks.cluster-c4xii6s2qytn.us-east-1.rds.amazonaws.com	Available	Writer	3306
epicbooks.cluster-ro-c4xii6s2qytn.us-east-1.rds.amazonaws.com	Available	Reader	3306

Manage IAM roles

- Select IAM roles to add to this cluster
  Add an existing IAM role to this cluster.
- Select a service to connect to this cluster
  Connect a service to this cluster by creating a new IAM role with permissions to access the service.

Choose an IAM role to add | Add role

Current IAM roles for this cluster (0)

Delete

Role	Status

Share DB cluster with other AWS accounts Info

Enables you to share your DB cluster with any AWS account or AWS Organizations through AWS Resource Access Manager (AWS RAM)

Add account ID to share this DB cluster

By sharing this DB cluster with other AWS accounts, you give them the ability to clone the cluster.

Add an account ID | Add account

You can add up to 15 accounts.

⚠ You can't share encrypted DB clusters with the default service key for RDS. Info

Accounts that this DB cluster is shared with (0)

Account ID	Status	Remove access from Resource shares

No account IDs added

Connected compute resources (1) Info

Actions

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

Resource identifier	Resource type	Availability Zone	VPC security group	Compute resource security group	Connected proxy
i-07cea6067484d5051	EC2 instance	us-east-1b	rds-ec2-1	ec2-rds-1	-

Proxies (0)

Actions | Create proxy

Filter by proxies

Proxy identifier	Status	Engine family

No proxies

You don't have any proxies.

Create proxy

RDS Data API Info

Enable the RDS Data API

With the RDS Data API enabled, you can run SQL queries against this database over HTTP. You can do so by using the CLI, an AWS SDK, or the RDS query editor. For information about pricing, see Amazon RDS pricing

Status of Data API

Disabled

```
sudo yum update -y
sudo yum install -y https://dev.mysql.com/get/mysql57-community-release-el7-11.noarch.rpm
sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2022
sudo yum-config-manager --disable mysql80-community
sudo yum-config-manager --enable mysql57-community
sudo yum install -y mysql-community-server
sudo systemctl status mysqld
```

```
[ec2-user@ip-172-31-28-251 theepicbook]$ mysql -h epicbooks.cluster-c4xii6s2qytn.us-east-1.rds.amazonaws.com -u admin -p

Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 84
Server version: 8.0.39 8bc99e28

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```
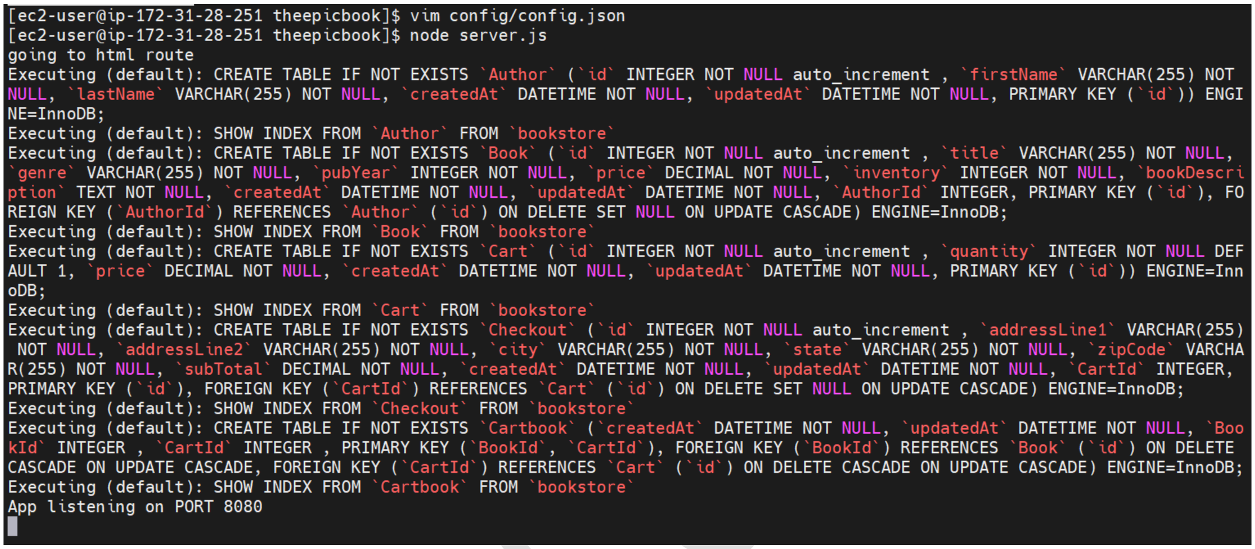

6. Setup MySQL database

```
mysql> CREATE DATABASE bookstore;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CREATE TABLE `bookstore`.`Author` (
    ->   `id` INT NOT NULL AUTO_INCREMENT,
    ->   `firstName` VARCHAR(45) NOT NULL,
    ->   `lastName` VARCHAR(45) NOT NULL,
    ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   PRIMARY KEY (`id`));
T NULL,
  `price` DECIMAL(13,2) NOT NULL,
  `inventory` INT NOT NULL,
  `bookDescription` TEXT NOT NULL,
  `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `AuthorId` INT NOT NULL,
  PRIMARY KEY (`id`),
  INDEX `AuthorId_idx` (`AuthorId` ASC),
  CONSTRAINT `AuthorId`
    FOREIGN KEY (`AuthorId`)
    REFERENCES `bookstore`.`Author` (`id`)
    ON DELETE NO ACTION
    ON UPDATE NO ACTION);



-- Create Table Cart
CREATE TABLE `bookstore`.`Cart` (
  `id` INT NOT NULL AUTO_INCREMENT,
  `quantity` INT NOT NULL,
  `price` DECIMAL(13,2) NOT NULL,
  `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`));Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- Create Table Book after Author because of FK constraint to Author Tbl
mysql> CREATE TABLE `bookstore`.`Book` (
    ->   `id` INT NOT NULL AUTO_INCREMENT,
    ->   `title` VARCHAR(255) NOT NULL,
    ->   `genre` VARCHAR(255) NOT NULL,
    ->   `pubYear` INT NOT NULL,
    ->   `price` DECIMAL(13,2) NOT NULL,
    ->   `inventory` INT NOT NULL,
    ->   `bookDescription` TEXT NOT NULL,
    ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   `AuthorId` INT NOT NULL,
    ->   PRIMARY KEY (`id`),
    ->   INDEX `AuthorId_idx` (`AuthorId` ASC),
    ->   CONSTRAINT `AuthorId`
    ->     FOREIGN KEY (`AuthorId`)
    ->     REFERENCES `bookstore`.`Author` (`id`)
    ->     ON DELETE NO ACTION
    ->     ON UPDATE NO ACTION);
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql>
mysql>
mysql> -- Create Table Cart
mysql> CREATE TABLE `bookstore`.`Cart` (
    ->   `id` INT NOT NULL AUTO_INCREMENT,
    ->   `quantity` INT NOT NULL,
    ->   `price` DECIMAL(13,2) NOT NULL,
    ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->   PRIMARY KEY (`id`));
Query OK, 0 rows affected (0.01 sec)

mysql> Show tables;
ERROR 1046 (3D000): No database selected
mysql> use bookstore
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+---------------------+
| Tables_in_bookstore |
+---------------------+
| Author              |
| Book                |
| Cart                |
+---------------------+
3 rows in set (0.00 sec)

mysql>
```

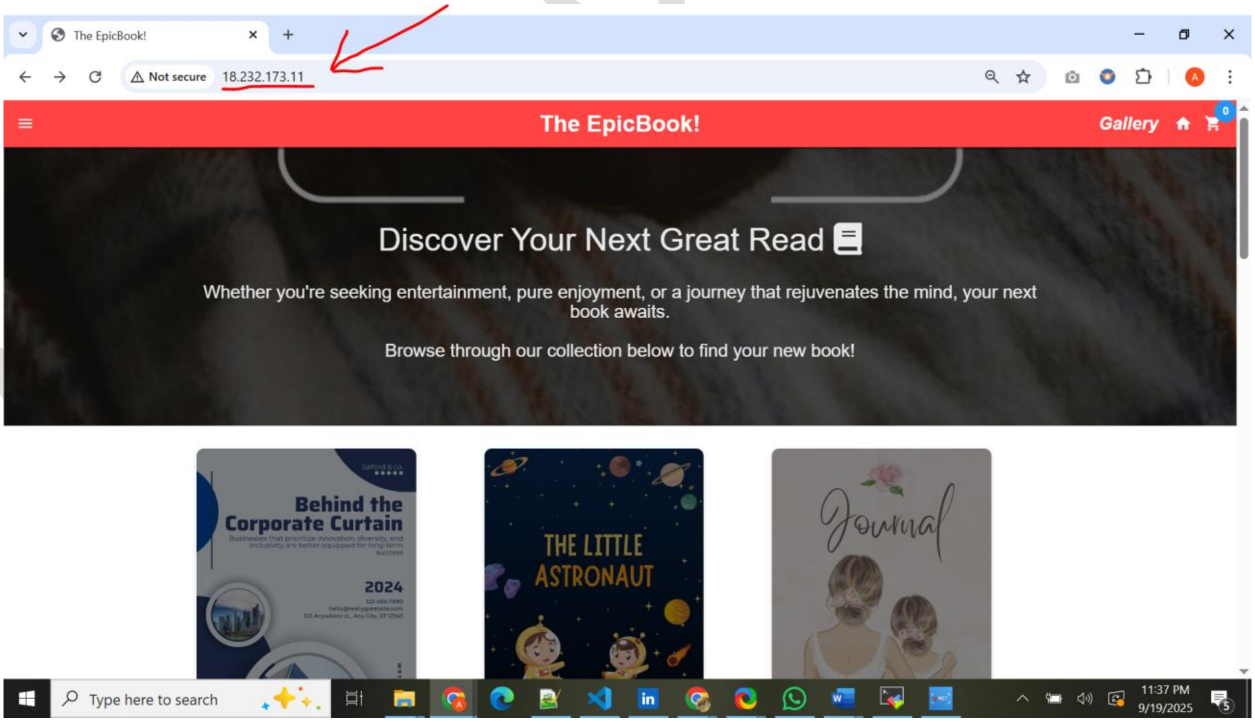
```
mysql> USE bookstore;
Database changed
mysql>
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Author(firstName,lastName,createdAt,updatedAt)
    -> VALUES
    -> ('Elin','Hildebrand',sysdate(),sysdate())
    -> ,('Brit','Bennett',sysdate(),sysdate())
    -> ,('John','Bolton',sysdate(),sysdate())
    -> ,('Robin','DiAngelo',sysdate(),sysdate())
    -> ,('Delia','Owens',sysdate(),sysdate())
    -> ,('John','Grisham',sysdate(),sysdate())
    -> ,('Ibram X.','Kendi',sysdate(),sysdate())
    -> ,('Celeste','Ng',sysdate(),sysdate())
    -> ,('Ijeoma','Oluo',sysdate(),sysdate())
    -> ,('Harper','Lee',sysdate(),sysdate())
    -> ,('L.M.','Montgomery',sysdate(),sysdate())
    -> ,('Jack','London',sysdate(),sysdate())
    -> ,('Jane','Austen',sysdate(),sysdate())
    -> ,('F. Scott','Fitzgerald',sysdate(),sysdate())
    -> ,('Emily','Bronte',sysdate(),sysdate())
    -> ,('Oscar','Wilde',sysdate(),sysdate())
    -> ,('George','Orwell',sysdate(),sysdate())
    -> ,('J.D.','Salinger',sysdate(),sysdate())
    -> ,('Shel','Silverstein',sysdate(),sysdate())
    -> ,('Maurice','Sendak',sysdate(),sysdate())
    -> ,('Dr.','Suess',sysdate(),sysdate())
    -> ,('Beatrix','Potter',sysdate(),sysdate())
    -> ,('C.S.','Lewis',sysdate(),sysdate())
    -> ,('E.B.','White',sysdate(),sysdate())
    -> ,('Eric','Carle',sysdate(),sysdate())
    -> ,('Michael','Ende',sysdate(),sysdate())
    -> ,('Howard','Zinn',sysdate(),sysdate())
    -> ,('Norton','Juster',sysdate(),sysdate())
    -> ,('Kurt','Vonnegut',sysdate(),sysdate())
    -> ,('Chinua','Achebe',sysdate(),sysdate())
    -> ,('Kazuo','Ishiguro',sysdate(),sysdate())
    -> ,('Art','Spiegelman',sysdate(),sysdate())
    -> ,('Margaret','Atwood',sysdate(),sysdate())
    -> ,('Douglas','Adams',sysdate(),sysdate())
    -> ,('David Foster','Wallace',sysdate(),sysdate())
    -> ,('Philip D.','Morgan',sysdate(),sysdate())
    -> ,('Hortense','Spillers',sysdate(),sysdate())
    -> ,('Stephen','Best',sysdate(),sysdate())
    -> ,('Anthony B.','Pinn',sysdate(),sysdate())
    -> ,('Harvey','Young',sysdate(),sysdate())
    -> ,('Michael A.','Gomez',sysdate(),sysdate())
    -> ,('Sarah','Haley',sysdate(),sysdate())
    -> ,('Daryl Michael','Scott',sysdate(),sysdate())
    -> ,('Cheryl','Harris',sysdate(),sysdate())
    -> ,('Frank','Herbert',sysdate(),sysdate())
    -> ,('Andrzej','Sapkowski',sysdate(),sysdate())
    -> ,('J.R.R.','Tolkein',sysdate(),sysdate())
    -> ,('J.K.','Rowling',sysdate(),sysdate())
    -> ,('Stephenie','Meyer',sysdate(),sysdate())
    -> ,('George R. R.','Martin',sysdate(),sysdate())
    -> ,('Diana','Gabaldon',sysdate(),sysdate())
    -> ,('Robert','Jordan',sysdate(),sysdate())
    -> ,('Cixin','Liu',sysdate(),sysdate());
Query OK, 53 rows affected (0.00 sec)
Records: 53  Duplicates: 0  Warnings: 0

mysql> -- Use JAWS_DB
mysql> USE bookstore;
Database changed
mysql>
mysql> /**
   /*> on your local machine, use the following:
   /*> USE bookstore_db;
   /*> **/
mysql>
mysql> -- Insert Books into Book Table
mysql> INSERT INTO Book(title,genre,pubYear,price,inventory,bookDescription,createdAt,updatedAt,AuthorId)
    ->
    -> -- Values are || Title | Genre | Published Year | Price | Inventory | Description | Created DtTm | Updated DtTm | Author ID ||
    -> VALUES
    ->  ('28 Summers','NYT',2020,'28',4,'A relationship that started in 1993 between Mallory Blessing and Jake McCloud comes to light while she is on her deathbed and his wife runs for president.',sysdate(),sysdate(),1)
    -> ,('The Vanishing Half','NYT',2020,'27',8,'The lives of twin sisters who run away from a Southern Black community at age 16 diverge as one returns and the other takes on a different racial identity but their fates intertwine.',sysdate(),sysdate(),2)
    -> ,('The Room Where it Happened','NYT',2020,'32.50',7,'The former national security advisor gives his account of the 17 months he spent working for President Trump.',sysdate(),sysdate(),3)
    -> ,('White Fragility','NYT',2018,'16',4,'Historical and cultural analyses on what causes defensive moves by white people and how this inhibits cross-racial dialogue.',sysdate(),sysdate(),4)
    -> ,('Where the Crawdads Sing','NYT',2018,'26',7,'In a quiet town on the North Carolina coast in 1969, a young woman who survived alone in the marsh becomes a murder suspect.',sysdate(),sysdate(),5)
    -> ,('Camino Winds','NYT',2020,'28.95',9,'The line between fact and fiction becomes blurred when an author of thrillers is found dead after a hurricane hits Camino Island.',sysdate(),sysdate(),6)
    -> ,('How to be an Antiracist','NYT',2019,'27',3,'A primer for creating a more just and equitable society through identifying and opposing racism.',sysdate(),sysdate(),7)
    -> ,('Little Fires Everywhere','NYT',2017,'17',5,'An artist with a mysterious past and a disregard for the status quo upends a quiet town outside Cleveland.',sysdate(),sysdate(),8)
    -> ,('So You Want To Talk About Race','NYT',2019,'16.99',2,'A look at the contemporary racial landscape of the United States',sysdate(),sysdate(),9)
    -> ,('To Kill A Mockingbird','Classic',1960,'15.99',10,'In "To Kill a Mockingbird", author Harper Lee uses memorable characters to explore civil rights and racism in the segregated Southern United States of the 1930s. Told through the eyes of Scout Finch, you learn about her father Atticus Finch, an attorney who hopelessly strives to prove the innocence of a black man unjustly accused of rape; and about Boo Radley, a mysterious neighbor who saves Scout and her brother Jem from being killed.',sysdate(),sysdate(),10)
    -> ,('Anne of Green Gables','Classic',1908,'5.99',8,'Anne of Green Gables is the story of a young orphan girl, Anne, who is mistakenly delivered to an older couple looking to adopt a boy to work on their farm in Avonlea on Prince Edward Island. After the couple meets Anne, they decide to keep her despite her not being of the appropriate gender.',sysdate(),sysdate(),11)
    -> ,('The Call of the Wild','Classic',1903,'12.95',6,'Based on Jack London''s experiences as a gold prospector in the Canadian wilderness and his ideas about nature and the struggle for existence, The Call of the Wild is a tale about unbreakable spirit and the fight for survival in the frozen Alaskan Klondike.',sysdate(),sysdate(),12)
    -> ,('Pride and Prejudice','Classic',1813,'5.00',9,'A classic of English literature , written with incisive wit and superb character delineation, it centres on the turbulent relationship between Elizabeth Bennet, the daughter of a country gentleman, and Fitzwilliam Darcy , a rich aristocratic landowner.',sysdate(),sysdate(),13)
    -> ,('The Great Gatsby','Classic',1925,'17.00',4,'The story is of the fabulously wealthy Jay Gatsby and his new love for the beautiful Daisy Buchanan, of lavish parties on Long Island at a time when The New York Times noted "gin was the national drink and sex the national obsession," it is an exquisitely crafted tale of America in the 1920s.',sysdate(),sysdate(),14)
    -> ,('Wuthering Heights','Classic',1847,'8.00',7,'Wuthering Heights is a classic gothic horror/romance novel written by Emily Bront and published in England in 1847. The book was originally published under the pen name Ellis Bell and Bronte died the year after it was released. When the book was first released it was considered controversial for its themes of violence and its challenge of Victorian ideals.',sysdate(),sysdate(),15)
    -> ,('The Picture of Dorian Gray','Classic',1890,'6.99',5,'In the stately London home of his aunt, Lady Brandon, the well-known artist Basil Hallward meets Dorian Gray. Dorian is a cultured, wealthy, and impossibly beautiful young man who immediately captures Basil''s artistic imagination. Dorian sits for several portraits, and Basil often depicts him as an ancient Greek hero or a mythological figure. The Picture of Dorian Gray is the story of one beautiful, innocent young man''s seduction, moral corruption, and eventual downfall.',sysdate(),sysdate(),16)
    -> ,('Animal Farm','Classic',1945,'18.50',3,'One of Orwell''s finest works, it is a political fable based on the events of Russia''s Bolshevik revolution and the betrayal of the cause by Joseph Stalin. The book concerns a group of barnyard animals who overthrow and chase off their exploitative human masters and set up an egalitarian society of their own.',sysdate(),sysdate(),17)
    -> ,('The Catcher in the Rye','Classic',1951,'9.99',2,'The novel details two days in the life of 16-year-old Holden Caulfield after he has been expelled from prep school. Confused and disillusioned, Holden searches for truth and rails against the phoniness of the adult world. He ends up exhausted and emotionally unstable.',sysdate(),sysdate(),18)
    -> ,('The Giving Tree','Childrens',1964,'17.99',8,'The book follows the lives of an apple tree and a boy, who develop a relationship with one another. The tree is very "giving" and the boy evolves into a "taking" teenager, man, then elderly man. Despite the fact that the boy ages in the story, the tree addresses the boy as "Boy" his entire life.',sysdate(),sysdate(),24)
    -> ,('Where the Wild Things Are','Childrens',1963,'8.55',3,'This story focuses on a young boy named Max who, after dressing in his wolf costume, wreaks such havoc through his household that he is sent to bed without his supper. Max''s bedroom undergoes a mysterious transformation into a jungle environment, and he winds up sailing to an island inhabited by malicious beasts known as the "Wild Things."',sysdate(),sysdate(),19)
    -> ,('The Cat in the Hat','Childrens',1957,'9.99',4,'The Cat in the Hat is a book where an eccentric stranger (who''s a cat!) comes into the house of two young children,Sally and Sam, who are having a very dull day. Their mother is out, and when the Cat comes in, he reassures the kids that their mother won''t mind him or his tricks!',sysdate(),sysdate(),20)
    -> ,('The Tale of Peter Rabbit','Childrens',1902,'7.99',6,'The widowed mother rabbit warns her four rabbit children, Flopsy, Mopsy, Cottontail and Peter not to enter the vegetable garden of a man named Mr. McGregor, whose wife, she tells them, put their father in a pie after he entered. Her triplets (Flopsy, Mopsy and Cottontail) obediently refrain from entering the garden, but Peter enters the garden to snack on some vegetables.',sysdate(),sysdate(),21)
    -> ,('The Lion, The Witch, and The Wardrobe','Childrens',1950,'8.99',5,'Peter, Susan, Edmund, and Lucy Pevensie are four siblings sent to live in the country with the eccentric Professor Kirke during World War II. The children explore the house on a rainy day and Lucy, the youngest, finds an enormous wardrobe. Lucy steps inside and finds herself in a strange, snowy wood.',sysdate(),sysdate(),22)
    -> ,('Charlotte''s Web','Childrens',1952,'9.99',9,'A masterpiece of American childrens literature, Charlotte''s Web is a fable by E.B. White about a runt of a pig named Wilbur, who is loved by a little girl and befriended by a very clever spider named Charlotte.',sysdate(),sysdate(),23)
    -> ,('Where the Sidewalk Ends','Childrens',1974,'19.99',7,'Where the Sidewalk Ends is a collection of quirky and imaginative poems by Shel Silverstein. The poems are accompanied by illustrations that capture the often silly, peculiar nature of Silverstein''s witty words.',sysdate(),sysdate(),24)
    -> ,('The Very Hungry Catepillar','Childrens',1969,'5.99',5,'This story follows the life of a growing caterpillar who begins as an egg. When he emerges from the egg he is very hungry. He starts to eat everything he can find, more and more each day.',sysdate(),sysdate(),25)
    -> ,('The Neverending Story','Childrens',1979,'8.99',3,'The book centres on a boy, Bastian, a strange child who is neglected by his father after the death of Bastian''s mother. While escaping from some bullies, Bastian bursts into the antiquarian book store of Carl Conrad Coreander , where he finds his interest held by a book called The Neverending Story',sysdate(),sysdate(),26)
    -> ,('A People''s History of the United States','Top 9',1980,'21.99',5,'A People''s History of the United States is the only volume to tell America''s story from the point of view of and in the words of America''s women, factory workers, African-Americans, Native Americans, the working poor, and immigrant laborers.',sysdate(),sysdate(),27)
    -> ,('The Phantom Tollbooth','Top 9',1961,'8.99',6,'The Phantom Tollbooth is the story of Milo, a very bored boy who comes home one day to find a magical tollbooth in his room. When Milo drives his car through the tollbooth gate, he finds himself in the Lands Beyond, a country inhabited by living language in the forms of animals, magicians, royalty, mountains, seas, and cities. From Tock the Watchdog to the listless region of The Doldrums, Milo shakes off boredom as he pursues the kidnapped Princesses Rhyme and Reason and restores peace to the Lands (currently in the clutches of the warring princes, Azaz of Dictionopolis and the Mathemagician of Digitopolis, along with a pack of demons).',sysdate(),sysdate(),28)
    -> ,('Slaughterhouse-Five','Top 9',1969,'17.00',10,'Slaughterhouse-Five, an American classic, is one of the worlds great antiwar books. Centering on the infamous firebombing of Dresden, Billy Pilgrims odyssey through time reflects the mythic journey of our own fractured lives as we search for meaning in what we fear most.',sysdate(),sysdate(),29)
    -> ,('Things Fall Apart','Top 9',1958,'13.00',11,'It is a classic narrative about Africa''s cataclysmic encounter with Europe as it establishes a colonial presence on the continent. Told through the fictional experiences of Okonkwo, a wealthy and fearless Igbo warrior of Umuofia in the late 1800s, Things Fall Apart explores one man''s futile resistance to the devaluing of his Igbo traditions by British political andreligious forces and his despair as his community capitulates to the powerful new order.',sysdate(),sysdate(),30)
    -> ,('Never Let Me Go','Top 9',2005,'16.00',4,'Friends Kathy, Tommy and Ruth grow up together at a seemingly idyllic boarding school in the English countryside. When they leave the school and the horrible truth of their true purpose is revealed to them, they must simultaneously confront deep-seated feelings of love, jealousy and betrayal that threaten to tear their friendship asunder.',sysdate(),sysdate(),31)
    -> ,('Maus','Top 9',1991,'35.00',8,'Maus depicts Spiegelman interviewing his father about his experiences as a Polish Jew and Holocaust survivor. The work employs postmodernist techniques and represents Jews as mice, Germans as cats, and Poles as pigs.',sysdate(),sysdate(),32)
    -> ,('The Handmaid''s Tale','Top 9',1985,'15.95',7,'The world of the narrator, Offred (from "Of Fred" women no longer have their own names), is chilling, but she is a magnificent survivor and chronicler, and the details of everything from mundane daily life to ritualized sex and violence to her reminiscences of the time before (our contemporary reality, as seen in the ''80s) are absolutely realistic.',sysdate(),sysdate(),33)
    -> ,('The Hitchhiker''s Guide to the Galaxy','Top 9',1980,'16.00',8,'A hapless hero with astonishing luck? Ill-tempered aliens hell-bent on destroying Earth? Pithy advice (e.g., "DON''T PANIC")? Check, check, and check and so much more. Even nonsci-fi geeks will be charmed by this hilarious and endlessly entertaining read, with (of course) sequels following.',sysdate(),sysdate(),34)
    -> ,('Infinite Jest','Top 9',1996,'20.00',9,'Infinite Jest is unique; it was bred in the optimism and new frontiersmanship of the dot-com 1990s but was simultaneously an early omen of where we are today. It looks into our present beyond what were only horizons when it was written: the tensions of a global economy, the opiate of on-demand entertainment, the near-impossible pursuit of greatness in a winner-take-all society. Tennis phenoms struggle in an absurdly demanding academy and recovering addicts search for something strong enough to help them through, all while a cadre of legless Quebecois assassins search for a movie so entertaining that they plan to use it as a weapon.',sysdate(),sysdate(),35)
    -> ,('Origins of American Slavery','Social Justice',2005,'17.00',3,'A concise but thorough overview of how American slavery fits into larger historical processes of the subjugation of black people, Philip Morgans description of slavery as an international institution looks at the institutions centrality in shaping global trends throughout the eighteenth and nineteenth centuries. By placing American slavery in a global context, Morgan explains how the expanding scale of capitalism and a long-standing perception of black inferiority converged to produce an instantiation of slavery that stands out as peculiarly heinous.',sysdate(),sysdate(),36)
    -> ,('Mama''s Baby, Papa''s Maybe: An American Grammar Book','Social Justice',1987,'13.00',5,'Hortense Spillers argues that discourses around blackness and gender during slavery continue to determine the ways in which black bodies are read in the United States. By focusing on black women, Spillers contends that we get it wrong when we only view slavery through the experience of enslaved people who identified as male. The machinations of enslavement as an institution fundamentally relied on fully disarticulating black women from the categories of woman and human.',sysdate(),sysdate(),37)
    -> ,('Neither Lost nor Found: Slavery and the Visual Archive','Social Justice',2011,'17.00',7,'Stephen Best argues that black people have often been understood as object rather than subjects. Best cites a case in which a photograph of two young black boys from the nineteenth century was advertised as rare at an estate sale. It was later discovered that copies of the photograph could be found at the Schomburg Center for Research in Black Culture and on eBay. Best addresses the paradox of the archival turn in studies of slavery: How can we claim to know that which cannot be known? That is, whether evidence can be found in the archive or not?',sysdate(),sysdate(),38)
    -> ,('Black Bodies in Pain and Ecstasy: Terror, Subjectivity, and the Nature of Black Religion','Social Justice',2003,'13.00',4,'Anthony Pinn argues that black religion, a capacious term that he purposefully deploys to reference a range of religious and spiritual practices beyond Christianity, plays a key role in African Americans struggle for what he calls complex subjectivity, a mode of being defined by ambiguity and multidimensionality. If dominant society defined black people by their corporeality during slavery, Pinn argues that black religion, which is an experience mediated by and through the black body, constitutes an important site of resistance against oppression. It contains an aesthetic, both in performance and style, that must be more prominently considered by the field of black religious studies.',sysdate(),sysdate(),39)
    -> ,('The Black Body as Souvenir in American Lynching','Social Justice',2005,'16.00',3,'Lynching, a phenomenon of extra-juridical violence used as a tool of social control, continues to be a lesser acknowledged practice in American history. Harvey Young analyzes what it meant for white participants in lynching spectacles to either steal or purchase the body parts of those who had been unjustly hanged, burned, castrated, and otherwise victimized and killed in public. Young asks what it means for white people to treat black bodies as souvenirs, fetish object, and remains.',sysdate(),sysdate(),40)
    -> ,('Of Du Bois and Diaspora: The Challenge of African American Studies','Social Justice',2004,'35.00',8,'Michael Gomez focuses on double consciousness, W. E. B. Du Bois term for the experience of African Americans who must simultaneously identify with blackness and Americanness. The nation signified in the latter term oppresses people based on the former term. Gomez makes a case for black intellectuals in the field of African American studies to more thoughtfully engage a larger diasporic approach with their work. Because the study of black people in America is a diasporic project about dispersal, loss, and community building, Gomez''s call for African American studies to take the diasporic turn seriously continues to influence the field.',sysdate(),sysdate(),41)
    -> ,('Like I Was a Man: Chain Gangs, Gender, and Domestic Carceral Sphere in Jim Crow Georgia','Social Justice',2013,'15.95',9,'Sarah Haley argues that the prison industrial complex has been an important site in which racialized conceptions of gender have been consolidated. Focusing on a comprehensive and historic prison reform act passed in Georgia in 1908 that forced imprisoned black women onto chain gangs and introduced a system of parole that compelled black women released from prison to become domestic servants in white homes, Haley argues that these reforms illustrate how black women were stripped of their gender. Incarcerated black women were obligated to perform both domestic labor that was gendered as female and hard physical labor that was gendered as male, but also, as a result of this un-gendering, they were legislated out of the category of the human.',sysdate(),sysdate(),42)
    -> ,('How Black Nationalism Became Sui Generis','Social Justice',2012,'16.00',2,'Daryl Michael Scott''s exploration of the vicissitudes of black nationalism that developed over the course of the early and mid-twentieth century negotiates a tension between black nationalism and other forms of nationalism. While nationalism on its own may be problematic, black nationalism has described everything from black separation and sovereignty to an imperium in imperio in which black people could self-determine without founding a new nation to a generic notion of racial solidarity among black people in a Pan-African context. The idea continues to have critical purchase in academia.',sysdate(),sysdate(),43)
    -> ,('Whiteness as Property','Social Justice',1993,'20.00',3,'Cheryl Harris explores the extent to which whiteness became a form of property that had to be protected by juridical and legislative means. As she argues, whiteness became valuable when white people could not be reduced to property under slavery. Given this, it is not only the cultural legacy of enslavement that has kept black people oppressed in the United States, but also a legal system that has continued to treat whiteness as the norm. That system actively excludes people of color from the purview of equal rights and protections while also affording economic benefits exclusively or disproportionately to white Americans.',sysdate(),sysdate(),44)
    -> ,('Dune','Fantasy',1965,'49.95',8,'Set on the desert planet Arrakis, Dune is the story of the boy Paul Atreides, heir to a noble family tasked with ruling an inhospitable world where the only thing of value is the spice melange, a drug capable of extending life and enhancing consciousness. Coveted across the known universe, melange is a prize worth killing for....',sysdate(),sysdate(),45)
    -> ,('The Witcher','Fantasy',1994,'39.99',8,'The Witcher is a fantasy series of novels and short stories written by Polish author Andrzej Sapkowski. The series revolves around the titular "witcher", Geralt of Rivia. In Sapkowski''s works, "witchers" are beast hunters who develop supernatural abilities at a young age to battle wild beasts and monsters.',sysdate(),sysdate(),46)
    -> ,('Lord of The Rings','Fantasy',1937,'35.96',8,'At 33, the age of adulthood among hobbits, Frodo Baggins receives a magic Ring of Invisibility from his uncle Bilbo. Frodo, a Christlike figure, learns that the ring has the power to control the entire world and, he discovers, to corrupt its owner. A fellowship of hobbits, elves, dwarfs, and men is formed to destroy the ring by casting it into the volcanic fires of the Crack of Doom, where it was forged. They are opposed on their harrowing mission by the evil Sauron and his Black Riders. The Lord of the Rings, together with The Hobbit, is considered by many to be the start of the genre known as high fantasy, and these works have had an enormous influence on that genre as a whole.',sysdate(),sysdate(),47)
    -> ,('Harry Potter','Fantasy',1998,'55.99',8,'Harry Potter, a boy who learns on his eleventh birthday that he is the orphaned son of two powerful wizards and possesses unique magical powers of his own. He is summoned from his life as an unwanted child to become a student at Hogwarts, an English boarding school for wizards. There, he meets several friends who become his closest allies and help him discover the truth about his parents'' mysterious deaths.',sysdate(),sysdate(),48)
    -> ,('Twilight','Fantasy',2005,'45.00',8,'The Twilight saga is a story about love. And vampires. And family. And abstinence. And racism. And the founding of the Mormon faith. And orphans, in a really weird way.',sysdate(),sysdate(),49)
    -> ,('A Song of Ice and Fire','Fantasy',1991,'50.00',8,'A Song of Ice and Fire is a series of epic fantasy novels by the American novelist and screenwriter George R. R. Martin. He began the first volume of the series, A Game of Thrones, in 1991, and it was published in 1996.',sysdate(),sysdate(),50)
    -> ,('Outlander','Fantasy',1991,'70.00',8,'Outlander is a series of historical romance science fiction novels by American author Diana Gabaldon. Gabaldon began the first volume of the series, Outlander, in the late 1980s, and it was published in 1991. She has published eight out of a planned ten volumes.',sysdate(),sysdate(),51)
    -> ,('Wheel of Time','Fantasy',1984,'32.00',8,'Written by Robert Jordan and completed posthumously by Brandon Sanderson, The Wheel of Time is one of the greatest epics of fantasy and a #1 internationally bestselling series. Taking place both in our past and our future, the saga tells the story of a man destined to face the Evil One and save the world or destroy it in the process.',sysdate(),sysdate(),52)
    -> ,('Three-Body Problem Trilogy','Fantasy',2008,'50.00',8,'From Hugo Award-winning author Cixin Lu, the most prolific and popular science fiction writer in the People''s Republic of China, the Remembrance of Earth''s Past series is an action-packed, thought-provoking saga of how humanity''s first contact with extraterrestrial life divided Earth between those welcoming the superior beings and those fighting the alien invasion.',sysdate(),sysdate(),53);
Query OK, 54 rows affected (0.01 sec)
Records: 54  Duplicates: 0  Warnings: 0

mysql>
```

7. Configure Database Connection in Node.js, Make changes in Config file to point to correct DB.
   node server.js



```
[ec2-user@ip-172-31-28-251 theepicbook]$ vim config/config.json
[ec2-user@ip-172-31-28-251 theepicbook]$ node server.js
going to html route
Executing (default): CREATE TABLE IF NOT EXISTS `Author` (`id` INTEGER NOT NULL auto_increment , `firstName` VARCHAR(255) NOT NULL, `lastName` VARCHAR(255) NOT NULL, `createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, PRIMARY KEY (`id`)) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Author` FROM `bookstore`
Executing (default): CREATE TABLE IF NOT EXISTS `Book` (`id` INTEGER NOT NULL auto_increment , `title` VARCHAR(255) NOT NULL, `genre` VARCHAR(255) NOT NULL, `pubYear` INTEGER NOT NULL, `price` DECIMAL NOT NULL, `inventory` INTEGER NOT NULL, `bookDescription` TEXT NOT NULL, `createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, `AuthorId` INTEGER, PRIMARY KEY (`id`), FOREIGN KEY (`AuthorId`) REFERENCES `Author` (`id`) ON DELETE SET NULL ON UPDATE CASCADE) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Book` FROM `bookstore`
Executing (default): CREATE TABLE IF NOT EXISTS `Cart` (`id` INTEGER NOT NULL auto_increment , `quantity` INTEGER NOT NULL DEFAULT 1, `price` DECIMAL NOT NULL, `createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, PRIMARY KEY (`id`)) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Cart` FROM `bookstore`
Executing (default): CREATE TABLE IF NOT EXISTS `Checkout` (`id` INTEGER NOT NULL auto_increment , `addressLine1` VARCHAR(255) NOT NULL, `addressLine2` VARCHAR(255) NOT NULL, `city` VARCHAR(255) NOT NULL, `state` VARCHAR(255) NOT NULL, `zipCode` VARCHAR(255) NOT NULL, `subTotal` DECIMAL NOT NULL, `createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, `CartId` INTEGER, PRIMARY KEY (`id`), FOREIGN KEY (`CartId`) REFERENCES `Cart` (`id`) ON DELETE SET NULL ON UPDATE CASCADE) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Checkout` FROM `bookstore`
Executing (default): CREATE TABLE IF NOT EXISTS `Cartbook` (`createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, `BookId` INTEGER , `CartId` INTEGER , PRIMARY KEY (`BookId`, `CartId`), FOREIGN KEY (`BookId`) REFERENCES `Book` (`id`) ON DELETE CASCADE ON UPDATE CASCADE, FOREIGN KEY (`CartId`) REFERENCES `Cart` (`id`) ON DELETE CASCADE ON UPDATE CASCADE) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Cartbook` FROM `bookstore`
App listening on PORT 8080
```

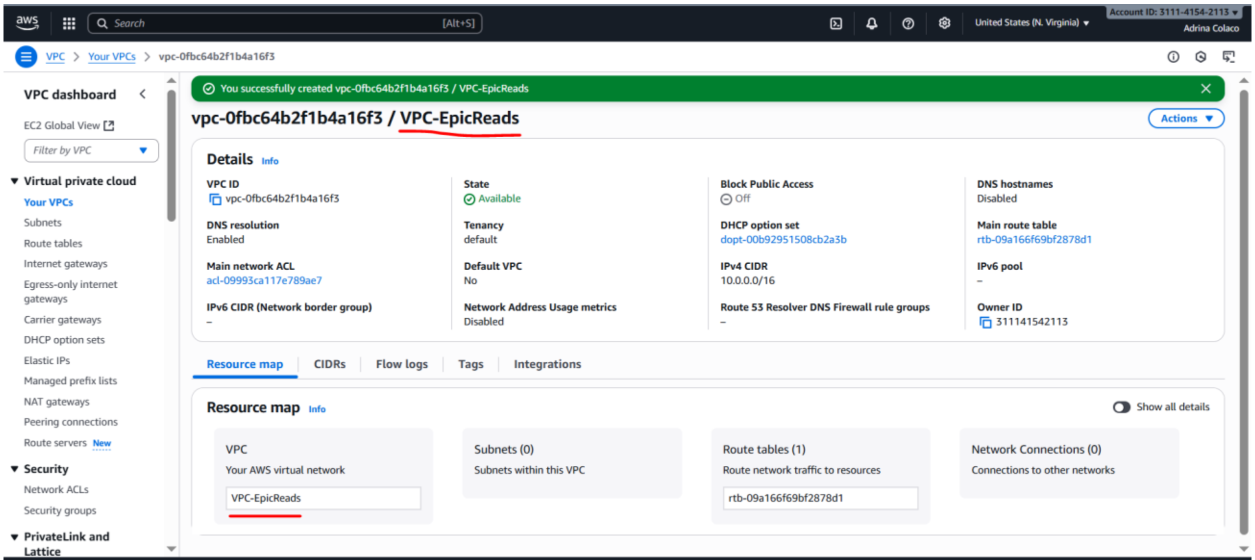
8. Verify the Setup



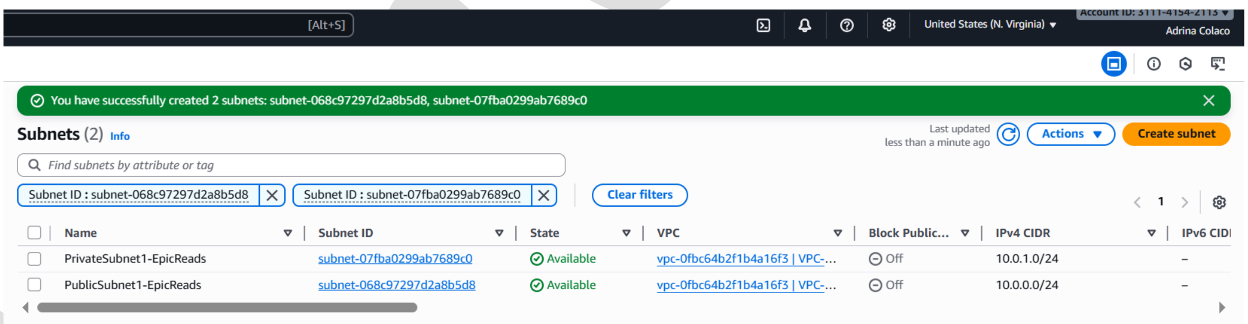
EpicBook_Monolith_App.mp4

Assignment 17 – Deploying a Two-Tier Application Architecture

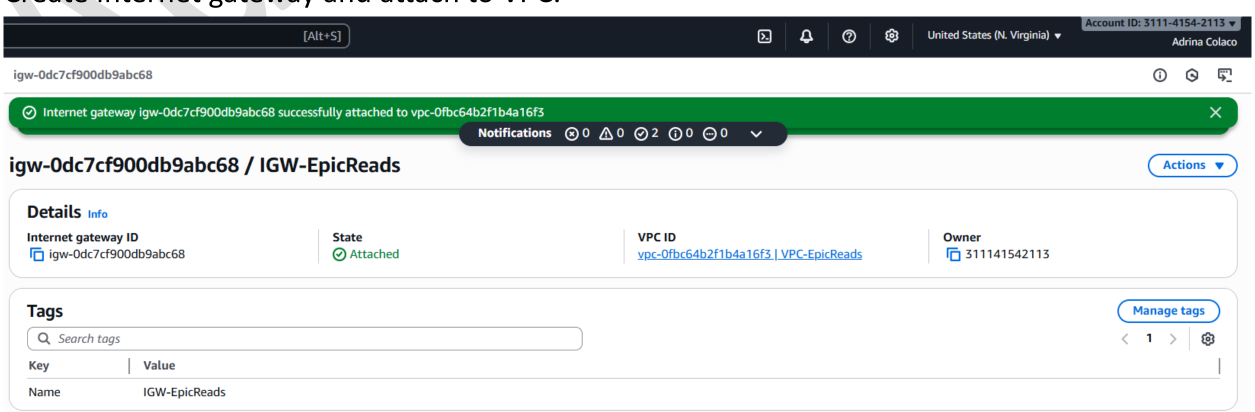
1. Create VPC



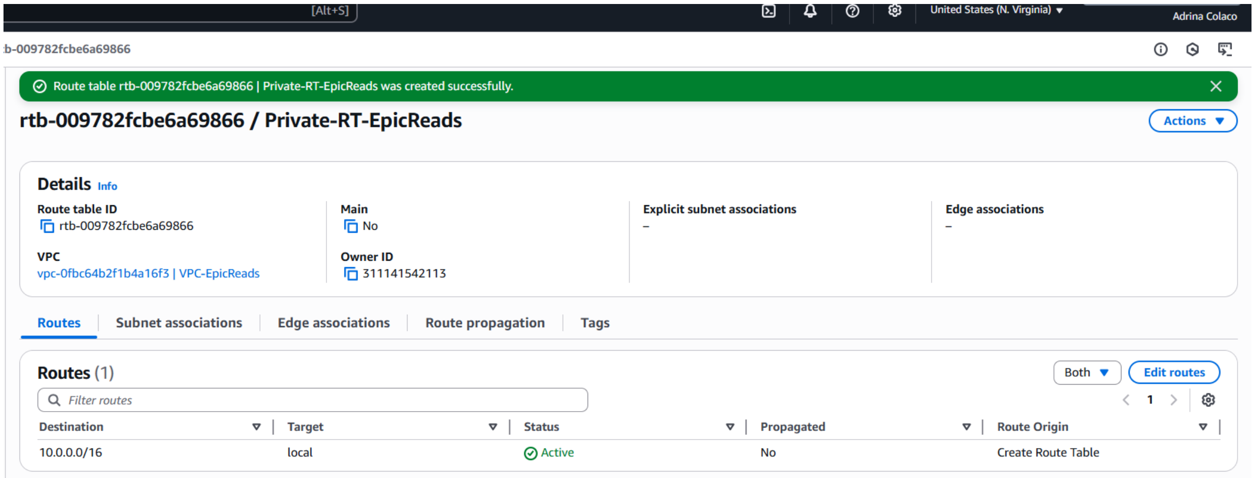
2. Create Subnet

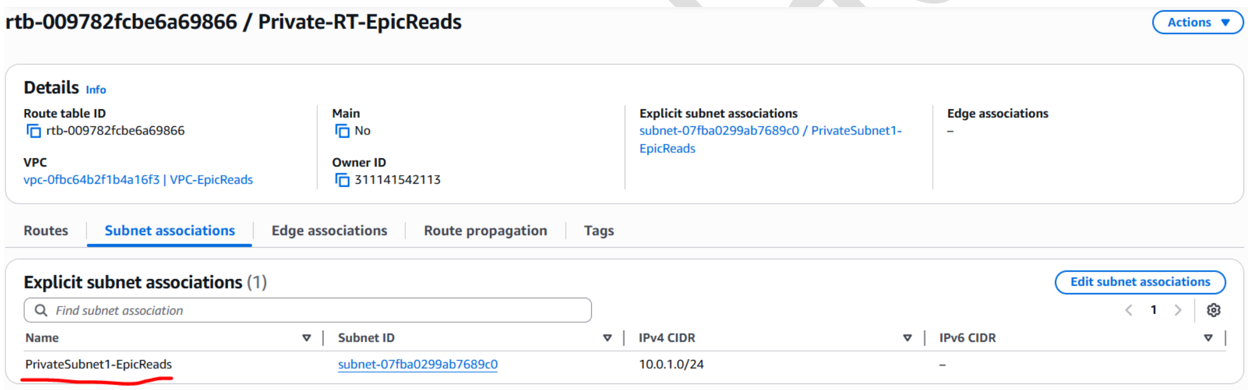


3. Create Internet gateway and attach to VPC.

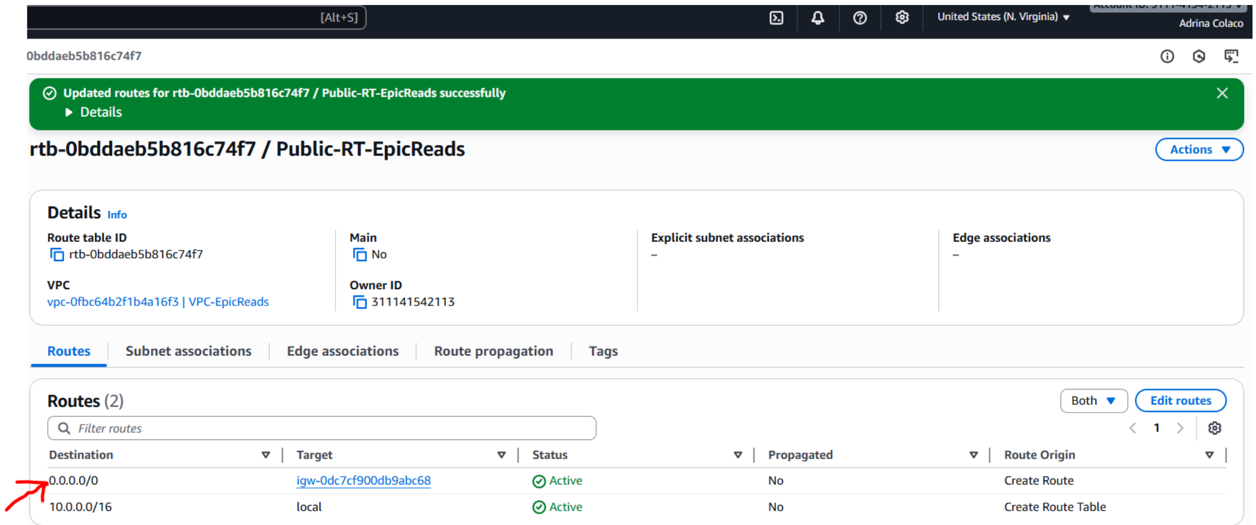


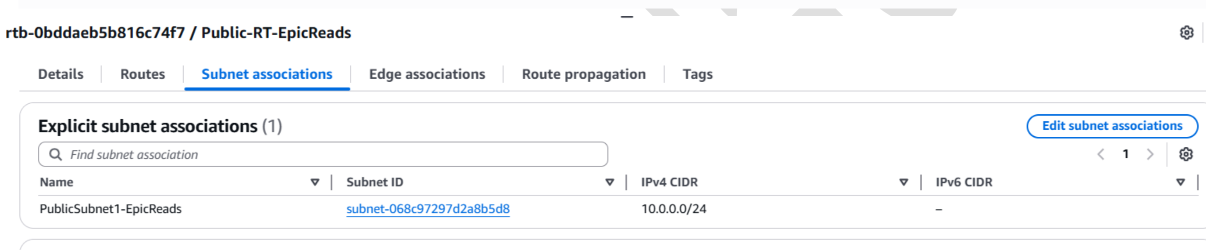
4. Create Private Route Table and Private Subnet in subnet associations



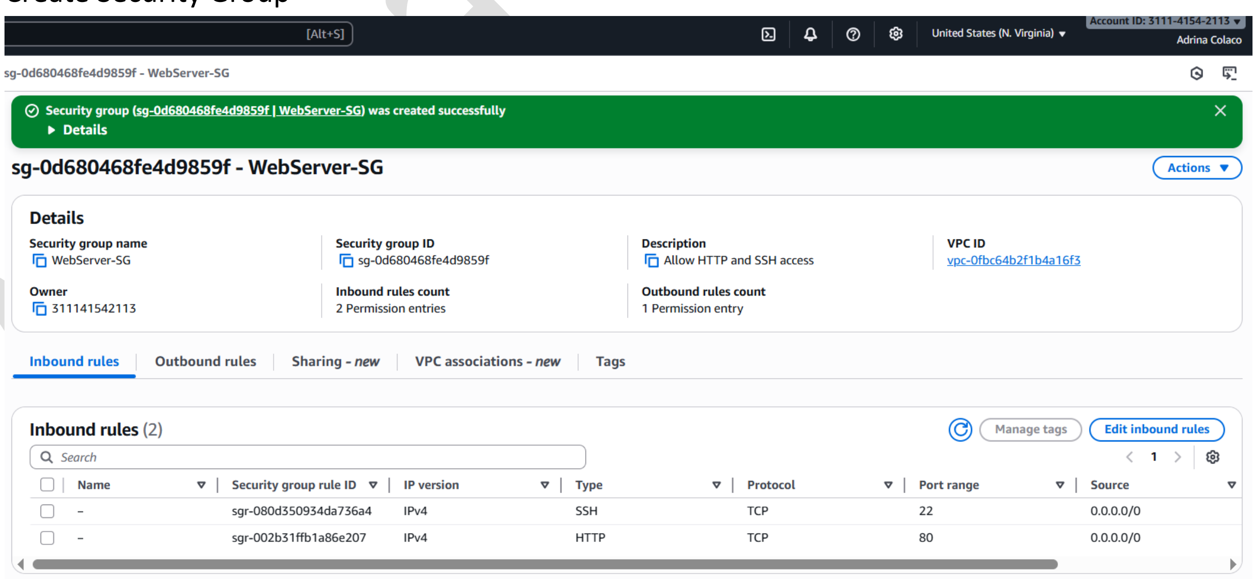


5. Create Public Route Table, add Internet Gateway in the Routes, and Public Subnet in subnet associations.

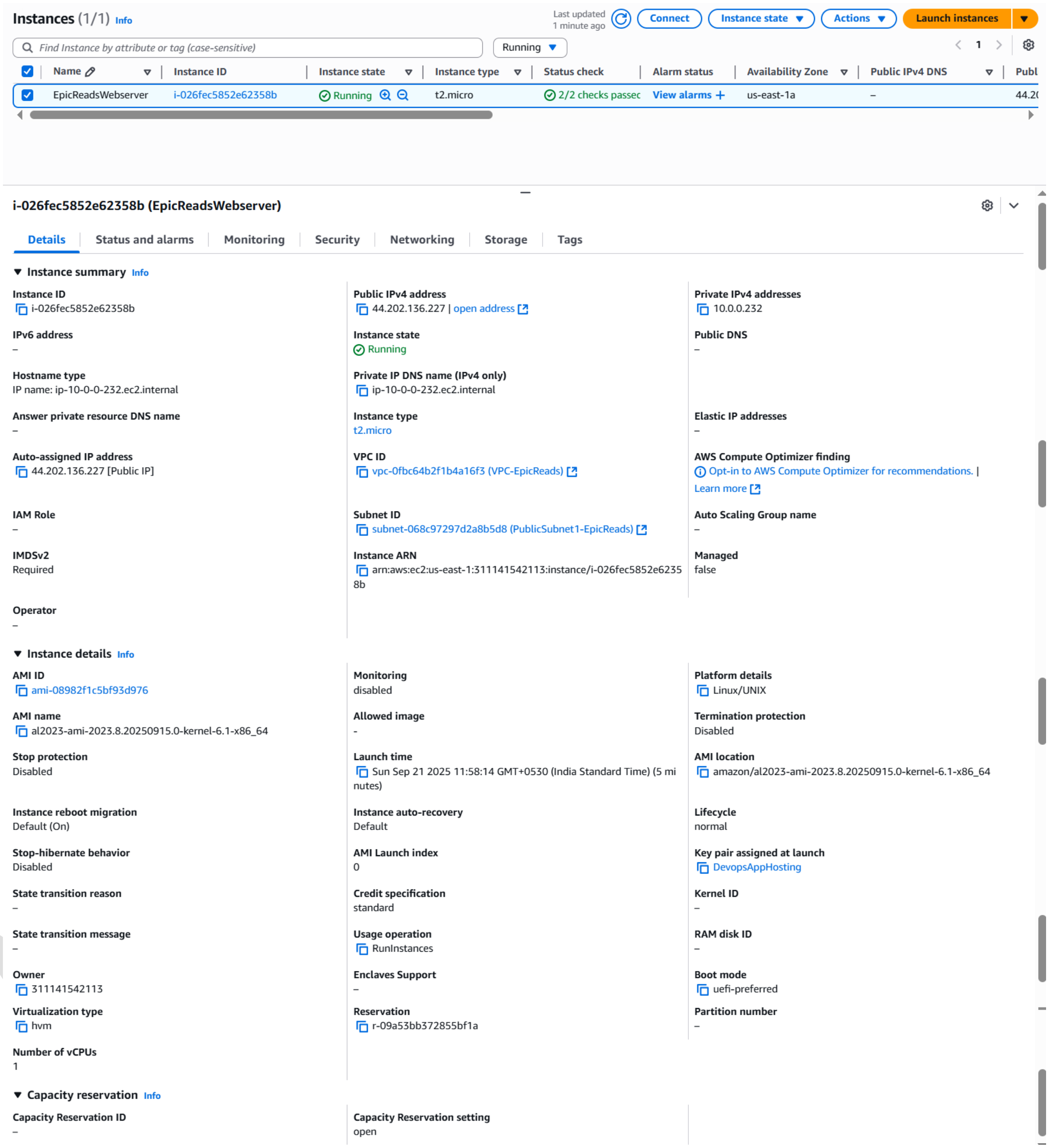




6. Create Security Group



7. Launch EC2 instance



8. Connect to EC2 and install HTTPD

sudo yum update

sudo yum install httpd -y

sudo systemctl start httpd

sudo systemctl status httpd

aws Search [Alt+S] United States (N. Virginia) ▾ Account ID: 3111-4154-2113 ▾ Adrina Colaco

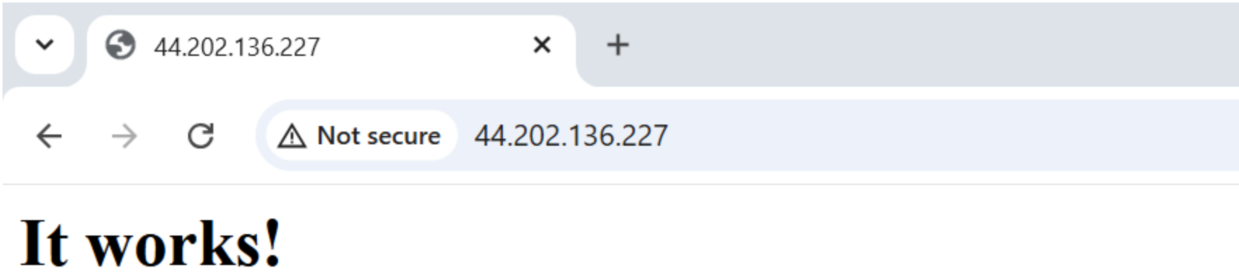
```
[ec2-user@ip-10-0-0-232 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                                   190 kB/s |  23 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:01:11 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
================================================================================================================
 Package                    Architecture        Version                          Repository               Size
================================================================================================================
Installing:
 httpd                      x86_64              2.4.65-1.amzn2023.0.1            amazonlinux              47 k
Installing dependencies:
 apr                        x86_64              1.7.5-1.amzn2023.0.4             amazonlinux             129 k
 apr-util                   x86_64              1.6.3-1.amzn2023.0.1             amazonlinux              98 k
 generic-logos-httpd        noarch              18.0.0-12.amzn2023.0.3           amazonlinux              19 k
 httpd-core                 x86_64              2.4.65-1.amzn2023.0.1            amazonlinux             1.4 M
 httpd-filesystem           noarch              2.4.65-1.amzn2023.0.1            amazonlinux              13 k
 httpd-tools                x86_64              2.4.65-1.amzn2023.0.1            amazonlinux              81 k
 libbrotli                  x86_64              1.0.9-4.amzn2023.0.2             amazonlinux             315 k
 mailcap                    noarch              2.1.49-3.amzn2023.0.3            amazonlinux              33 k
Installing weak dependencies:
 apr-util-openssl           x86_64              1.6.3-1.amzn2023.0.1             amazonlinux              17 k
 mod_http2                  x86_64              2.0.27-1.amzn2023.0.3            amazonlinux             166 k
 mod_lua                    x86_64              2.4.65-1.amzn2023.0.1            amazonlinux              60 k

Transaction Summary
================================================================================================================
Install  12 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Downloading Packages:
(1/12): apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64.rpm                        517 kB/s |  17 kB     00:00
(2/12): apr-util-1.6.3-1.amzn2023.0.1.x86_64.rpm                                2.1 MB/s |  98 kB     00:00
(3/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                     2.6 MB/s | 129 kB     00:00
(4/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                   803 kB/s |  19 kB     00:00
(5/12): httpd-2.4.65-1.amzn2023.0.1.x86_64.rpm                                  2.2 MB/s |  47 kB     00:00
(6/12): httpd-filesystem-2.4.65-1.amzn2023.0.1.noarch.rpm                       593 kB/s |  13 kB     00:00
(7/12): httpd-tools-2.4.65-1.amzn2023.0.1.x86_64.rpm                            3.7 MB/s |  81 kB     00:00
(8/12): httpd-core-2.4.65-1.amzn2023.0.1.x86_64.rpm                              27 MB/s | 1.4 MB     00:00
(9/12): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                                1.4 MB/s |  33 kB     00:00
(10/12): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                              8.7 MB/s | 315 kB     00:00
(11/12): mod_http2-2.0.27-1.amzn2023.0.3.x86_64.rpm                             5.6 MB/s | 166 kB     00:00
(12/12): mod_lua-2.4.65-1.amzn2023.0.1.x86_64.rpm                               2.9 MB/s |  60 kB     00:00
----------------------------------------------------------------------------------------------------------------
Total                                                                            14 MB/s | 2.3 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                       1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                                                       1/12
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64                                          2/12
  Installing       : apr-util-1.6.3-1.amzn2023.0.1.x86_64                                                  3/12
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                  4/12
  Installing       : httpd-tools-2.4.65-1.amzn2023.0.1.x86_64                                              5/12
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                 6/12
  Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.1.noarch                                         7/12
  Installing       : httpd-filesystem-2.4.65-1.amzn2023.0.1.noarch                                         7/12
  Installing       : httpd-core-2.4.65-1.amzn2023.0.1.x86_64                                               8/12
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                9/12
  Installing       : mod_lua-2.4.65-1.amzn2023.0.1.x86_64                                                 10/12
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                    11/12
  Installing       : httpd-2.4.65-1.amzn2023.0.1.x86_64                                                   12/12
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.1.x86_64                                                   12/12
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                                       1/12
  Verifying        : apr-util-1.6.3-1.amzn2023.0.1.x86_64                                                  2/12
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64                                          3/12
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                     4/12
  Verifying        : httpd-2.4.65-1.amzn2023.0.1.x86_64                                                    5/12
  Verifying        : httpd-core-2.4.65-1.amzn2023.0.1.x86_64                                               6/12
  Verifying        : httpd-filesystem-2.4.65-1.amzn2023.0.1.noarch                                         7/12
  Verifying        : httpd-tools-2.4.65-1.amzn2023.0.1.x86_64                                              8/12
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                 9/12
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                 10/12
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                               11/12
  Verifying        : mod_lua-2.4.65-1.amzn2023.0.1.x86_64                                                 12/12

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.1.x86_64       apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-2.4.65-1.amzn2023.0.1.x86_64       httpd-core-2.4.65-1.amzn2023.0.1.x86_64    httpd-filesystem-2.4.65-1.amzn2023.0.1.noarch     httpd-tools-2.4.65-1.amzn2023.0.1.x86_64
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64    mailcap-2.1.49-3.amzn2023.0.3.noarch       mod_http2-2.0.27-1.amzn2023.0.3.x86_64            mod_lua-2.4.65-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-0-232 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Sun 2025-09-21 06:37:08 UTC; 11s ago
       Docs: man:httpd.service(8)
   Main PID: 21835 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1111)
     Memory: 13.0M
        CPU: 72ms
     CGroup: /system.slice/httpd.service
             ├─21835 /usr/sbin/httpd -DFOREGROUND
             ├─21925 /usr/sbin/httpd -DFOREGROUND
             ├─21929 /usr/sbin/httpd -DFOREGROUND
             ├─21930 /usr/sbin/httpd -DFOREGROUND
             └─21931 /usr/sbin/httpd -DFOREGROUND

Sep 21 06:37:08 ip-10-0-0-232.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 21 06:37:08 ip-10-0-0-232.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 21 06:37:08 ip-10-0-0-232.ec2.internal httpd[21835]: Server configured, listening on: port 80
[ec2-user@ip-10-0-0-232 ~]$
```

i-026fec5852e62358b (EpicReadsWebserver)

PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

CloudShell Feedback Privacy Terms Cookie preferences

9. Open Public URL of EC2 and see if HTTPD works.



10. Install PHP and MySQL

    ```
    sudo dnf update -y
    sudo dnf install -y php php-mysqlnd
    sudo dnf install -y mariadb105-server
    ```

aws Search [Alt+S] United States (N. Virginia) ▾ Account ID: 3111-4154-2113 ▾ Adrina Colaco

```
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf update -y
Last metadata expiration check: 0:04:03 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf install -y php php-mysqlnd
Last metadata expiration check: 0:04:17 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
================================================================================
 Package              Architecture   Version                   Repository   Size
================================================================================
Installing:
 php8.4               x86_64         8.4.10-1.amzn2023.0.1     amazonlinux  17 k
 php8.4-mysqlnd       x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 156 k
Installing dependencies:
 libsodium            x86_64         1.0.19-4.amzn2023         amazonlinux 176 k
 libxslt              x86_64         1.1.43-1.amzn2023.0.2     amazonlinux 183 k
 nginx-filesystem     noarch         1:1.28.0-1.amzn2023.0.2   amazonlinux 9.6 k
 php8.4-cli           x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 3.8 M
 php8.4-common        x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 800 k
 php8.4-pdo           x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 100 k
 php8.4-process       x86_64         8.4.10-1.amzn2023.0.1     amazonlinux  52 k
 php8.4-xml           x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 966 k
Installing weak dependencies:
 php8.4-fpm           x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 2.0 M
 php8.4-mbstring      x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 541 k
 php8.4-opcache       x86_64         8.4.10-1.amzn2023.0.1     amazonlinux 495 k
 php8.4-sodium        x86_64         8.4.10-1.amzn2023.0.1     amazonlinux  48 k

Transaction Summary
================================================================================
Install  14 Packages

Total download size: 9.3 M
Installed size: 42 M
Downloading Packages:
(1/14): libsodium-1.0.19-4.amzn2023.x86_64.rpm            4.6 MB/s | 176 kB     00:00
(2/14): nginx-filesystem-1.28.0-1.amzn2023.0.2.noarch.rpm 244 kB/s | 9.6 kB     00:00
(3/14): libxslt-1.1.43-1.amzn2023.0.2.x86_64.rpm          4.0 MB/s | 183 kB     00:00
(4/14): php8.4-8.4.10-1.amzn2023.0.1.x86_64.rpm           756 kB/s |  17 kB     00:00
(5/14): php8.4-common-8.4.10-1.amzn2023.0.1.x86_64.rpm     23 MB/s | 800 kB     00:00
(6/14): php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64.rpm        60 MB/s | 3.8 MB     00:00
(7/14): php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64.rpm   13 MB/s | 541 kB     00:00
(8/14): php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64.rpm        28 MB/s | 2.0 MB     00:00
(9/14): php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64.rpm   4.8 MB/s | 156 kB     00:00
(10/14): php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64.rpm   14 MB/s | 495 kB     00:00
(11/14): php8.4-process-8.4.10-1.amzn2023.0.1.x86_64.rpm  2.5 MB/s |  52 kB     00:00
(12/14): php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64.rpm      3.1 MB/s | 100 kB     00:00
(13/14): php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64.rpm   2.0 MB/s |  48 kB     00:00
(14/14): php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64.rpm       29 MB/s | 966 kB     00:00
--------------------------------------------------------------------------------
Total                                                      41 MB/s | 9.3 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : php8.4-common-8.4.10-1.amzn2023.0.1.x86_64            1/14
  Installing       : php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64               2/14
  Installing       : php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64               3/14
  Installing       : php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64          4/14
  Installing       : php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64           5/14
  Installing       : php8.4-process-8.4.10-1.amzn2023.0.1.x86_64           6/14
  Running scriptlet: nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch       7/14
  Installing       : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch       7/14
  Installing       : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64               8/14
  Running scriptlet: php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64               8/14
  Installing       : libxslt-1.1.43-1.amzn2023.0.2.x86_64                  9/14
  Installing       : php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64              10/14
  Installing       : libsodium-1.0.19-4.amzn2023.x86_64                   11/14
  Installing       : php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64           12/14
  Installing       : php8.4-8.4.10-1.amzn2023.0.1.x86_64                  13/14
  Installing       : php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64          14/14
  Running scriptlet: php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64          14/14
  Verifying        : libsodium-1.0.19-4.amzn2023.x86_64                    1/14
  Verifying        : libxslt-1.1.43-1.amzn2023.0.2.x86_64                  2/14
  Verifying        : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch       3/14
  Verifying        : php8.4-8.4.10-1.amzn2023.0.1.x86_64                   4/14
  Verifying        : php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64               5/14
  Verifying        : php8.4-common-8.4.10-1.amzn2023.0.1.x86_64            6/14
  Verifying        : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64               7/14
  Verifying        : php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64          8/14
  Verifying        : php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64           9/14
  Verifying        : php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64          10/14
  Verifying        : php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64              11/14
  Verifying        : php8.4-process-8.4.10-1.amzn2023.0.1.x86_64          12/14
  Verifying        : php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64           13/14
  Verifying        : php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64              14/14

Installed:
  libsodium-1.0.19-4.amzn2023.x86_64          libxslt-1.1.43-1.amzn2023.0.2.x86_64          nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch  php8.4-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64     php8.4-common-8.4.10-1.amzn2023.0.1.x86_64    php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64          php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64 php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64   php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64          php8.4-process-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64  php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf install -y mariadb105-server
Last metadata expiration check: 0:05:16 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
================================================================================
 Package                          Architecture  Version                     Repository   Size
================================================================================
Installing:
 mariadb105-server                x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  10 M
Installing dependencies:
 mariadb-connector-c              x86_64        3.3.10-1.amzn2023.0.1       amazonlinux 211 k
 mariadb-connector-c-config       noarch        3.3.10-1.amzn2023.0.1       amazonlinux 9.9 k
 mariadb105                       x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux 1.5 M
 mariadb105-common                x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  28 k
 mariadb105-errmsg                x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux 212 k
 mysql-selinux                    noarch        1.0.4-2.amzn2023.0.3        amazonlinux  36 k
 perl-B                           x86_64        1.80-477.amzn2023.0.7       amazonlinux 177 k
 perl-DBD-MariaDB                 x86_64        1.22-1.amzn2023.0.4         amazonlinux 153 k
 perl-DBI                         x86_64        1.643-7.amzn2023.0.3        amazonlinux 700 k
 perl-Data-Dumper                 x86_64        2.174-460.amzn2023.0.2      amazonlinux  55 k
 perl-File-Copy                   noarch        2.34-477.amzn2023.0.7       amazonlinux  20 k
 perl-FileHandle                  noarch        2.03-477.amzn2023.0.7       amazonlinux  15 k
 perl-Math-BigInt                 noarch        1:1.9998.39-2.amzn2023.0.2  amazonlinux 202 k
 perl-Math-BigRat                 noarch        0.2624-500.amzn2023.0.2     amazonlinux  42 k
 perl-Math-Complex                noarch        1.59-477.amzn2023.0.7       amazonlinux  46 k
 perl-Sys-Hostname                x86_64        1.23-477.amzn2023.0.7       amazonlinux  16 k
 perl-base                        noarch        2.27-477.amzn2023.0.7       amazonlinux  16 k
Installing weak dependencies:
 mariadb105-backup                x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux 6.0 M
 mariadb105-cracklib-password-check x86_64      3:10.5.29-1.amzn2023.0.1    amazonlinux  13 k
 mariadb105-gssapi-server         x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  15 k
 mariadb105-server-utils          x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux 207 k

Transaction Summary
================================================================================
Install  22 Packages

Total download size: 20 M
Installed size: 117 M
Downloading Packages:
(1/22): mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch.rpm           291 kB/s | 9.9 kB     00:00
(2/22): mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64.rpm                  4.9 MB/s | 211 kB     00:00
(3/22): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm                           24 MB/s | 1.5 MB     00:00
(4/22): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm                   1.3 MB/s |  28 kB     00:00
(5/22): mariadb105-cracklib-password-check-10.5.29-1.amzn2023.0.1.x86_64.rpm  610 kB/s |  13 kB     00:00
(6/22): mariadb105-errmsg-10.5.29-1.amzn2023.0.1.x86_64.rpm                   7.9 MB/s | 212 kB     00:00
(7/22): mariadb105-backup-10.5.29-1.amzn2023.0.1.x86_64.rpm                    62 MB/s | 6.0 MB     00:00
(8/22): mariadb105-gssapi-server-10.5.29-1.amzn2023.0.1.x86_64.rpm            292 kB/s |  15 kB     00:00
(9/22): mariadb105-server-utils-10.5.29-1.amzn2023.0.1.x86_64.rpm             8.7 MB/s | 207 kB     00:00
(10/22): mysql-selinux-1.0.4-2.amzn2023.0.3.noarch.rpm                        1.5 MB/s |  36 kB     00:00
(11/22): perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64.rpm                      5.4 MB/s | 153 kB     00:00
(12/22): perl-B-1.80-477.amzn2023.0.7.x86_64.rpm                              5.0 MB/s | 177 kB     00:00
(13/22): mariadb105-server-10.5.29-1.amzn2023.0.1.x86_64.rpm                   62 MB/s |  10 MB     00:00
(14/22): perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64.rpm                   842 kB/s |  55 kB     00:00
(15/22): perl-DBI-1.643-7.amzn2023.0.3.x86_64.rpm                             8.9 MB/s | 700 kB     00:00
(16/22): perl-File-Copy-2.34-477.amzn2023.0.7.noarch.rpm                      988 kB/s |  20 kB     00:00
(17/22): perl-FileHandle-2.03-477.amzn2023.0.7.noarch.rpm                     719 kB/s |  15 kB     00:00
(18/22): perl-Math-BigInt-1.9998.39-2.amzn2023.0.2.noarch.rpm                 6.5 MB/s | 202 kB     00:00
(19/22): perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch.rpm                  1.7 MB/s |  42 kB     00:00
(20/22): perl-Math-Complex-1.59-477.amzn2023.0.7.noarch.rpm                   2.0 MB/s |  46 kB     00:00
(21/22): perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64.rpm                   693 kB/s |  16 kB     00:00
(22/22): perl-base-2.27-477.amzn2023.0.7.noarch.rpm                           722 kB/s |  16 kB     00:00
--------------------------------------------------------------------------------
Total                                                                          55 MB/s |  20 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch   1/22
  Installing       : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64         2/22
  Installing       : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64          3/22
  Installing       : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64            4/22
  Installing       : perl-Math-Complex-1.59-477.amzn2023.0.7.noarch            5/22
  Installing       : perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch           6/22
  Installing       : perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch        7/22
  Installing       : mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64         8/22
  Installing       : perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64            9/22
  Installing       : perl-B-1.80-477.amzn2023.0.7.x86_64                      10/22
  Installing       : perl-base-2.27-477.amzn2023.0.7.noarch                   11/22
  Installing       : perl-FileHandle-2.03-477.amzn2023.0.7.noarch             12/22
  Installing       : perl-DBI-1.643-7.amzn2023.0.3.x86_64                     13/22
  Installing       : perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64              14/22
  Installing       : perl-File-Copy-2.34-477.amzn2023.0.7.noarch              15/22
  Running scriptlet: mysql-selinux-1.0.4-2.amzn2023.0.3.noarch                16/22
```

i-026fec5852e62358b (EpicReadsWebserver)

PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

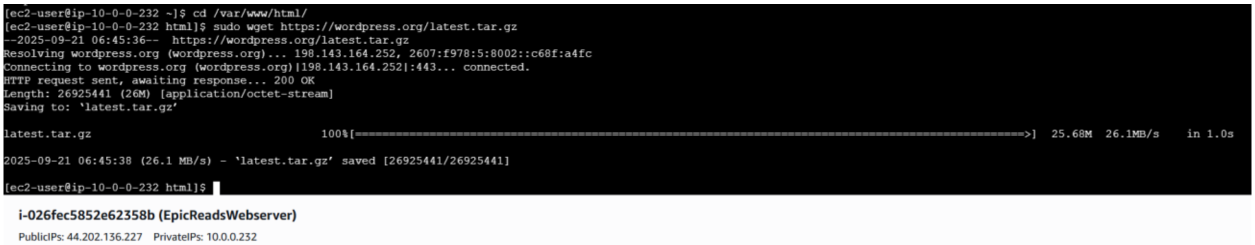
Search [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco

```
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf update -y
Last metadata expiration check: 0:04:03 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf install -y php php-mysqlnd
Last metadata expiration check: 0:04:17 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
================================================================================
 Package                 Architecture   Version                  Repository     Size
================================================================================
Installing:
 php8.4                  x86_64         8.4.10-1.amzn2023.0.1    amazonlinux    17 k
 php8.4-mysqlnd          x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   156 k
Installing dependencies:
 libsodium               x86_64         1.0.19-4.amzn2023        amazonlinux   176 k
 libxslt                 x86_64         1.1.43-1.amzn2023.0.2    amazonlinux   183 k
 nginx-filesystem        noarch         1:1.28.0-1.amzn2023.0.2  amazonlinux   9.6 k
 php8.4-cli              x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   3.8 M
 php8.4-common           x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   800 k
 php8.4-pdo              x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   100 k
 php8.4-process          x86_64         8.4.10-1.amzn2023.0.1    amazonlinux    52 k
 php8.4-xml              x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   966 k
Installing weak dependencies:
 php8.4-fpm              x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   2.0 M
 php8.4-mbstring         x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   541 k
 php8.4-opcache          x86_64         8.4.10-1.amzn2023.0.1    amazonlinux   495 k
 php8.4-sodium           x86_64         8.4.10-1.amzn2023.0.1    amazonlinux    48 k

Transaction Summary
================================================================================
Install  14 Packages

Total download size: 9.3 M
Installed size: 42 M
Downloading Packages:
(1/14): libsodium-1.0.19-4.amzn2023.x86_64.rpm                   4.6 MB/s | 176 kB     00:00
(2/14): nginx-filesystem-1.28.0-1.amzn2023.0.2.noarch.rpm        244 kB/s | 9.6 kB     00:00
(3/14): libxslt-1.1.43-1.amzn2023.0.2.x86_64.rpm                 4.0 MB/s | 183 kB     00:00
(4/14): php8.4-8.4.10-1.amzn2023.0.1.x86_64.rpm                  756 kB/s |  17 kB     00:00
(5/14): php8.4-common-8.4.10-1.amzn2023.0.1.x86_64.rpm            23 MB/s | 800 kB     00:00
(6/14): php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64.rpm               60 MB/s | 3.8 MB     00:00
(7/14): php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64.rpm          13 MB/s | 541 kB     00:00
(8/14): php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64.rpm               28 MB/s | 2.0 MB     00:00
(9/14): php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64.rpm          4.8 MB/s | 156 kB     00:00
(10/14): php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64.rpm          14 MB/s | 495 kB     00:00
(11/14): php8.4-process-8.4.10-1.amzn2023.0.1.x86_64.rpm         2.5 MB/s |  52 kB     00:00
(12/14): php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64.rpm             3.1 MB/s | 100 kB     00:00
(13/14): php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64.rpm          2.0 MB/s |  48 kB     00:00
(14/14): php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64.rpm              29 MB/s | 966 kB     00:00
--------------------------------------------------------------------------------
Total                                                             41 MB/s | 9.3 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                          1/1
  Installing       : php8.4-common-8.4.10-1.amzn2023.0.1.x86_64               1/14
  Installing       : php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64                  2/14
  Installing       : php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64                  3/14
  Installing       : php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64             4/14
  Installing       : php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64              5/14
  Installing       : php8.4-process-8.4.10-1.amzn2023.0.1.x86_64              6/14
  Running scriptlet: nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch          7/14
  Installing       : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch          7/14
  Installing       : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                  8/14
  Running scriptlet: php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                  8/14
  Installing       : libxslt-1.1.43-1.amzn2023.0.2.x86_64                     9/14
  Installing       : php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64                 10/14
  Installing       : libsodium-1.0.19-4.amzn2023.x86_64                      11/14
  Installing       : php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64              12/14
  Installing       : php8.4-8.4.10-1.amzn2023.0.1.x86_64                     13/14
  Installing       : php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64             14/14
  Running scriptlet: php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64             14/14
  Verifying        : libsodium-1.0.19-4.amzn2023.x86_64                       1/14
  Verifying        : libxslt-1.1.43-1.amzn2023.0.2.x86_64                     2/14
  Verifying        : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch          3/14
  Verifying        : php8.4-8.4.10-1.amzn2023.0.1.x86_64                      4/14
  Verifying        : php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64                  5/14
  Verifying        : php8.4-common-8.4.10-1.amzn2023.0.1.x86_64               6/14
  Verifying        : php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64                  7/14
  Verifying        : php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64             8/14
  Verifying        : php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64              9/14
  Verifying        : php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64             10/14
  Verifying        : php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64                 11/14
  Verifying        : php8.4-process-8.4.10-1.amzn2023.0.1.x86_64             12/14
  Verifying        : php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64              13/14
  Verifying        : php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64                 14/14

Installed:
  libsodium-1.0.19-4.amzn2023.x86_64          libxslt-1.1.43-1.amzn2023.0.2.x86_64          nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch  php8.4-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-cli-8.4.10-1.amzn2023.0.1.x86_64     php8.4-common-8.4.10-1.amzn2023.0.1.x86_64    php8.4-fpm-8.4.10-1.amzn2023.0.1.x86_64          php8.4-mbstring-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-mysqlnd-8.4.10-1.amzn2023.0.1.x86_64 php8.4-opcache-8.4.10-1.amzn2023.0.1.x86_64   php8.4-pdo-8.4.10-1.amzn2023.0.1.x86_64          php8.4-process-8.4.10-1.amzn2023.0.1.x86_64
  php8.4-sodium-8.4.10-1.amzn2023.0.1.x86_64  php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-0-232 ~]$ sudo dnf install -y mariadb105-server
Last metadata expiration check: 0:05:16 ago on Sun Sep 21 06:35:43 2025.
Dependencies resolved.
================================================================================
 Package                            Architecture  Version                     Repository    Size
================================================================================
Installing:
 mariadb105-server                  x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux   10 M
Installing dependencies:
 mariadb-connector-c                x86_64        3.3.10-1.amzn2023.0.1       amazonlinux  211 k
 mariadb-connector-c-config         noarch        3.3.10-1.amzn2023.0.1       amazonlinux  9.9 k
 mariadb105                         x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  1.5 M
 mariadb105-common                  x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux   28 k
 mariadb105-errmsg                  x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  212 k
 mysql-selinux                      noarch        1.0.4-2.amzn2023.0.3        amazonlinux   36 k
 perl-B                             x86_64        1.80-477.amzn2023.0.7       amazonlinux  177 k
 perl-DBD-MariaDB                   x86_64        1.22-1.amzn2023.0.4         amazonlinux  153 k
 perl-DBI                           x86_64        1.643-7.amzn2023.0.3        amazonlinux  700 k
 perl-Data-Dumper                   x86_64        2.174-460.amzn2023.0.2      amazonlinux   55 k
 perl-File-Copy                     noarch        2.34-477.amzn2023.0.7       amazonlinux   20 k
 perl-FileHandle                    noarch        2.03-477.amzn2023.0.7       amazonlinux   15 k
 perl-Math-BigInt                   noarch        1:1.9998.39-2.amzn2023.0.2  amazonlinux  202 k
 perl-Math-BigRat                   noarch        0.2624-500.amzn2023.0.2     amazonlinux   42 k
 perl-Math-Complex                  noarch        1.59-477.amzn2023.0.7       amazonlinux   46 k
 perl-Sys-Hostname                  x86_64        1.23-477.amzn2023.0.7       amazonlinux   16 k
 perl-base                          noarch        2.27-477.amzn2023.0.7       amazonlinux   16 k
Installing weak dependencies:
 mariadb105-backup                  x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  6.0 M
 mariadb105-cracklib-password-check x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux   13 k
 mariadb105-gssapi-server           x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux   15 k
 mariadb105-server-utils            x86_64        3:10.5.29-1.amzn2023.0.1    amazonlinux  207 k

Transaction Summary
================================================================================
Install  22 Packages

Total download size: 20 M
Installed size: 117 M
Downloading Packages:
(1/22): mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch.rpm              291 kB/s | 9.9 kB     00:00
(2/22): mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64.rpm                     4.9 MB/s | 211 kB     00:00
(3/22): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm                              24 MB/s | 1.5 MB     00:00
(4/22): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm                      1.3 MB/s |  28 kB     00:00
(5/22): mariadb105-cracklib-password-check-10.5.29-1.amzn2023.0.1.x86_64.rpm     610 kB/s |  13 kB     00:00
(6/22): mariadb105-errmsg-10.5.29-1.amzn2023.0.1.x86_64.rpm                      7.9 MB/s | 212 kB     00:00
(7/22): mariadb105-backup-10.5.29-1.amzn2023.0.1.x86_64.rpm                       62 MB/s | 6.0 MB     00:00
(8/22): mariadb105-gssapi-server-10.5.29-1.amzn2023.0.1.x86_64.rpm               292 kB/s |  15 kB     00:00
(9/22): mariadb105-server-utils-10.5.29-1.amzn2023.0.1.x86_64.rpm                8.7 MB/s | 207 kB     00:00
(10/22): mysql-selinux-1.0.4-2.amzn2023.0.3.noarch.rpm                           1.5 MB/s |  36 kB     00:00
(11/22): perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64.rpm                         5.4 MB/s | 153 kB     00:00
(12/22): perl-B-1.80-477.amzn2023.0.7.x86_64.rpm                                 5.0 MB/s | 177 kB     00:00
(13/22): mariadb105-server-10.5.29-1.amzn2023.0.1.x86_64.rpm                      62 MB/s |  10 MB     00:00
(14/22): perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64.rpm                      842 kB/s |  55 kB     00:00
(15/22): perl-DBI-1.643-7.amzn2023.0.3.x86_64.rpm                                8.9 MB/s | 700 kB     00:00
(16/22): perl-File-Copy-2.34-477.amzn2023.0.7.noarch.rpm                         988 kB/s |  20 kB     00:00
(17/22): perl-FileHandle-2.03-477.amzn2023.0.7.noarch.rpm                        719 kB/s |  15 kB     00:00
(18/22): perl-Math-BigInt-1.9998.39-2.amzn2023.0.2.noarch.rpm                    6.5 MB/s | 202 kB     00:00
(19/22): perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch.rpm                     1.7 MB/s |  42 kB     00:00
(20/22): perl-Math-Complex-1.59-477.amzn2023.0.7.noarch.rpm                      2.0 MB/s |  46 kB     00:00
(21/22): perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64.rpm                      693 kB/s |  16 kB     00:00
(22/22): perl-base-2.27-477.amzn2023.0.7.noarch.rpm                              722 kB/s |  16 kB     00:00
--------------------------------------------------------------------------------
Total                                                                             55 MB/s |  20 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                          1/1
  Installing       : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch  1/22
  Installing       : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64        2/22
  Installing       : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64         3/22
  Installing       : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64           4/22
  Installing       : perl-Math-Complex-1.59-477.amzn2023.0.7.noarch           5/22
  Installing       : perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch          6/22
  Installing       : perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch       7/22
  Installing       : mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64        8/22
  Installing       : perl-Data-Dumper-2.174-460.amzn2023.0.2.x86_64           9/22
  Installing       : perl-B-1.80-477.amzn2023.0.7.x86_64                     10/22
  Installing       : perl-base-2.27-477.amzn2023.0.7.noarch                  11/22
  Installing       : perl-FileHandle-2.03-477.amzn2023.0.7.noarch            12/22
  Installing       : perl-DBI-1.643-7.amzn2023.0.3.x86_64                    13/22
  Installing       : perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64             14/22
  Installing       : perl-File-Copy-2.34-477.amzn2023.0.7.noarch             15/22
  Running scriptlet: mysql-selinux-1.0.4-2.amzn2023.0.3.noarch               16/22
```

i-026fec5852e62358b (EpicReadsWebserver)

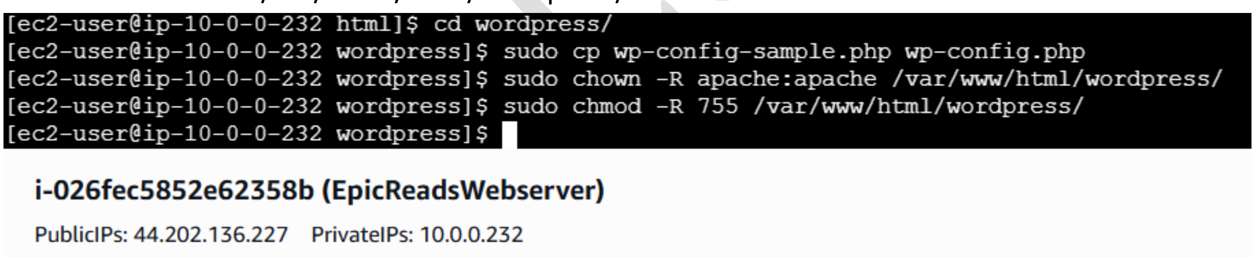
PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

11. Goto /var/www/html folder, download WordPress and unzip it.
    cd /var/www/html/
    sudo wget https://wordpress.org/latest.tar.gz
    sudo tar -xzvf latest.tar.gz

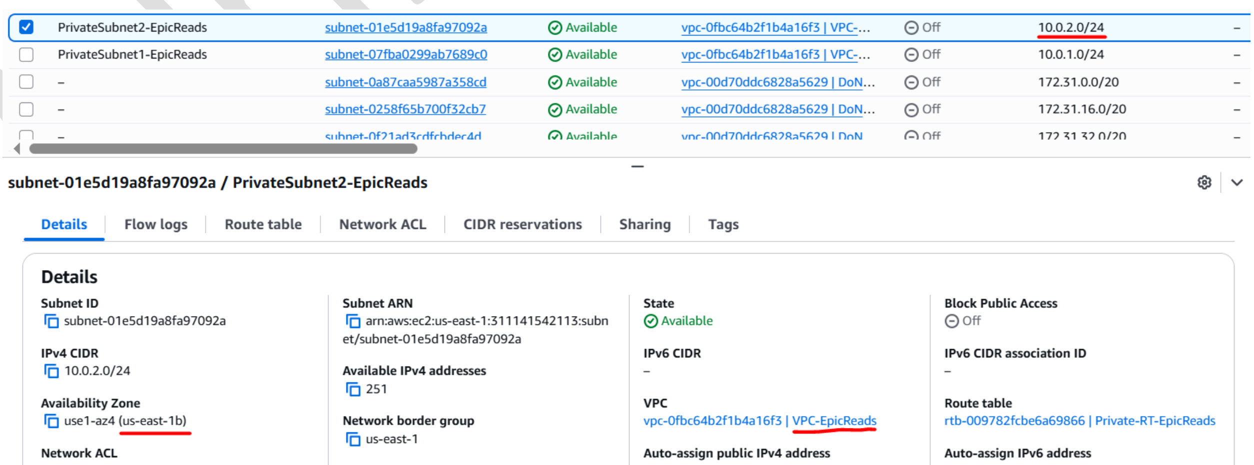


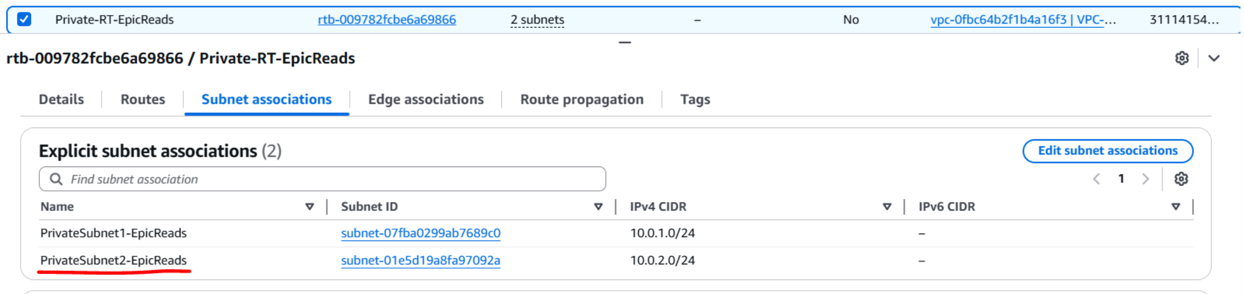
12. Go into wordpress directory,
    cd wordpress/
    # Create configuration file(Using sample config file)
    sudo cp wp-config-sample.php wp-config.php

    #Adjust permissions for wordpress files
    sudo chown -R apache:apache /var/www/html/wordpress/
    sudo chmod -R 755 /var/www/html/wordpress/

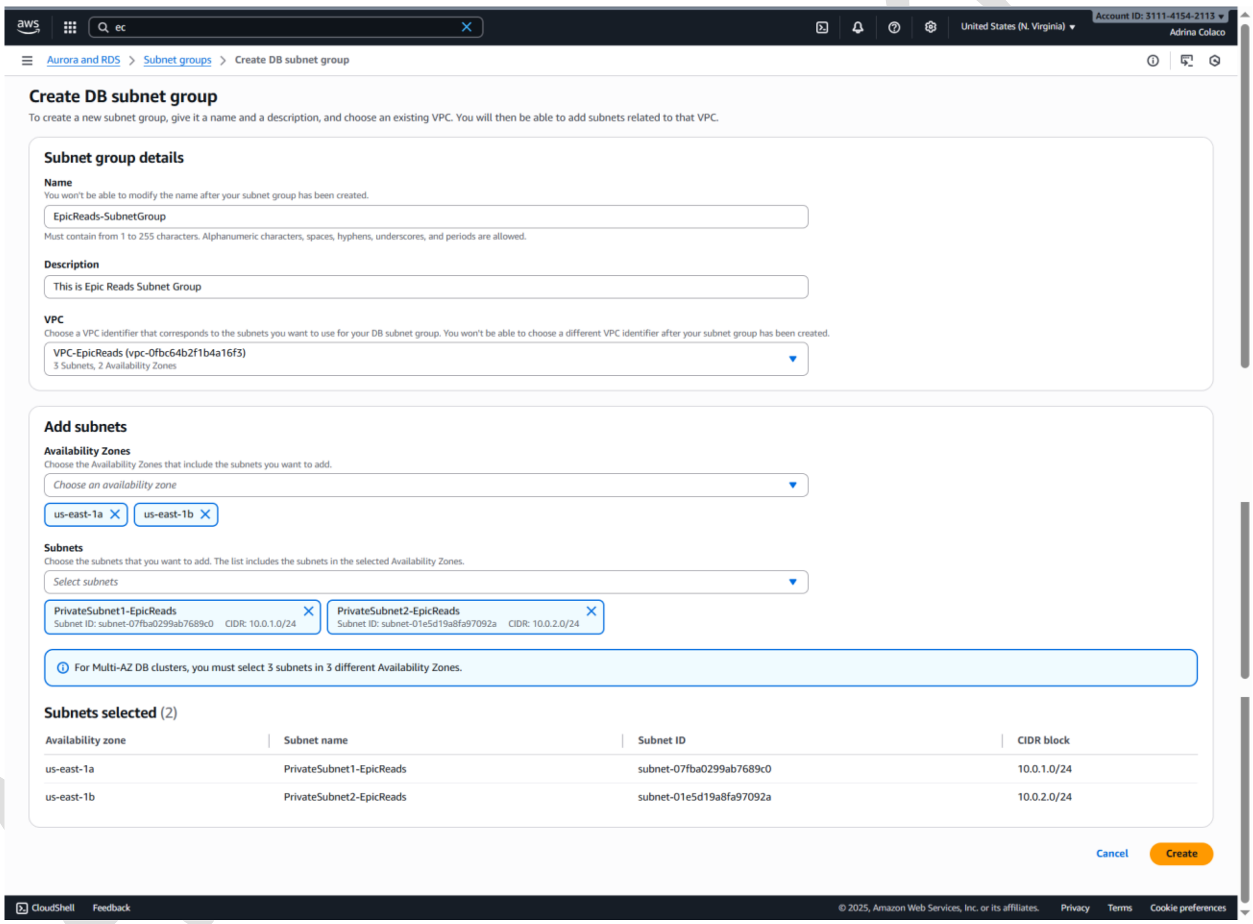


13. Create another Private Subnet and associate it in the Private Route table.





14. Create an RDS subnet group



15. Create MySQL database

database-1

Successfully created database database-1

RDS has generated your database master password during the database creation and it will be displayed in the connection details. The only way to view your master password is to choose **View connection details** during database creation. You can modify your DB instance to create a new password at any time.

You can use settings from database-1 to simplify configuration of **suggested database add-ons** while we finish creating your DB for you.

View connection details

database-1

Modify | Actions

Summary

DB identifier	Status	Role	Engine	Recommendations
database-1	Available	Instance	MySQL Community	
CPU	**Class**	**Current activity**	**Region & AZ**	
-	db.t4g.micro		us-east-1b	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations - *new*

Connectivity & security

Endpoint & port

Endpoint
database-1.c4xii6s2qytn.us-east-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-1b

VPC
VPC-EpicReads (vpc-0fbc64b2f1b4a16f3)

Subnet group
epicreads-subnetgroup

Subnets
subnet-01e5d19a8fa97092a
subnet-07fba0299ab7689c0

Network type
IPv4

Security

VPC security groups
EpicReads-DBSG (sg-0a9dbbb9a50062d30)
Active

Publicly accessible
No

Certificate authority Info
rds-ca-rsa2048-g1

Certificate authority date
May 26, 2061, 05:04 (UTC+05:30)

DB instance certificate expiration date
September 21, 2026, 12:55 (UTC+05:30)

Connected compute resources (0) Info

Actions

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

Resource identifier	Resource type	Availability Zone	VPC security group	Compute resource security group	Connected proxy

No connected compute resources

No connected compute resources that were created automatically to display.

Set up EC2 connection | Set up Lambda connection

Proxies (0)

Actions | Create proxy

Filter by proxies

Proxy identifier	Status	Engine family

No proxies

You don't have any proxies.

Create proxy

Security group rules (2)

Filter by Security group rules

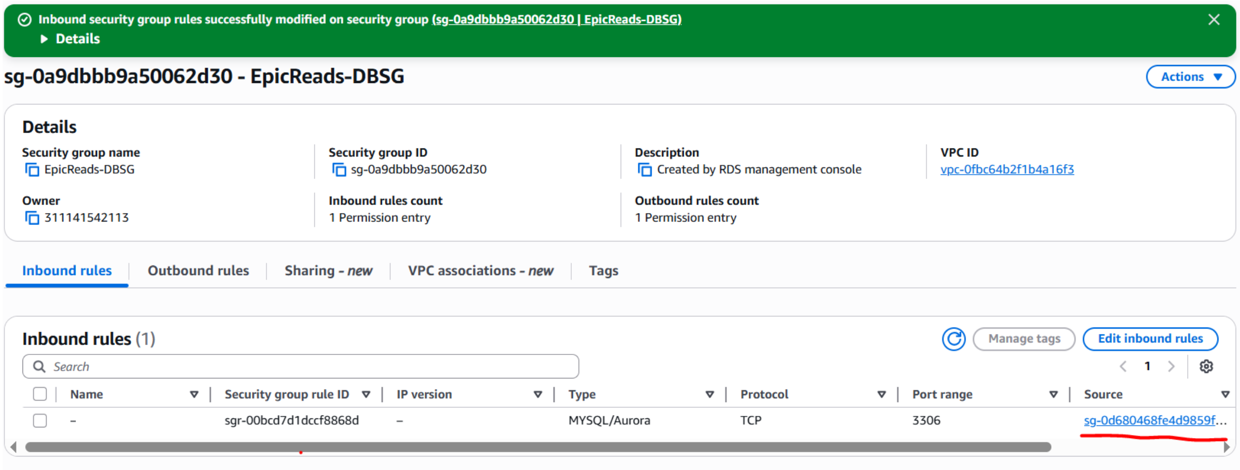
Security group	Type	Rule
EpicReads-DBSG (sg-0a9dbbb9a50062d30)	CIDR/IP - Inbound	223.177.210.62/32
EpicReads-DBSG (sg-0a9dbbb9a50062d30)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

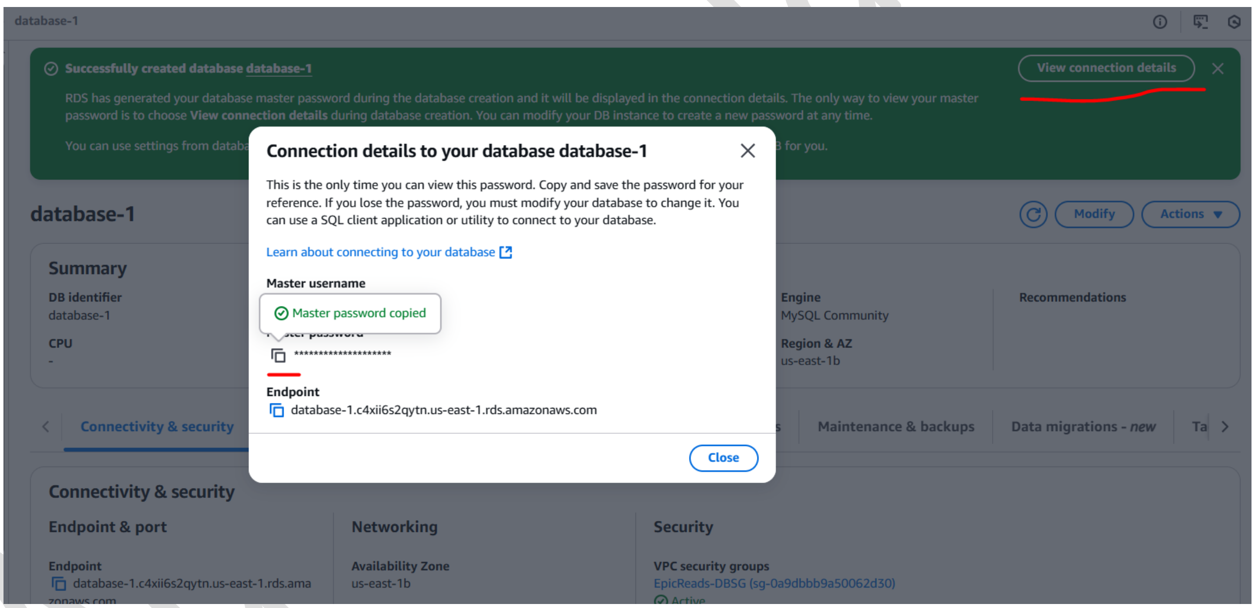
Filter by Replication

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
database-1	Instance	us-east-1b	-	-	-

16. Edit the Database security group to give access to web server security group.

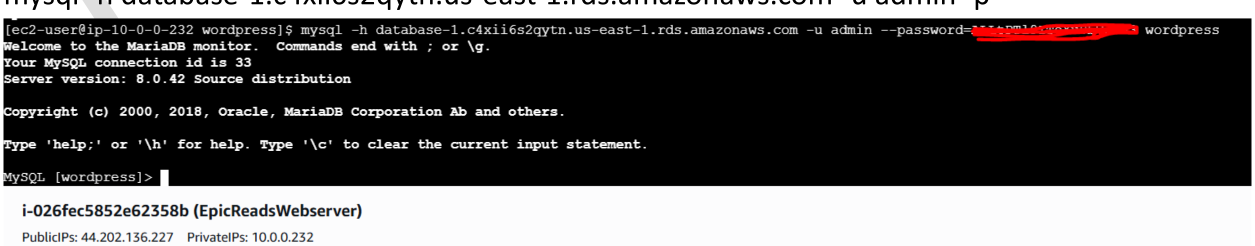


17. Test if you can access the DB from webserver



Option 1:

mysql -h database-1.c4xii6s2qytn.us-east-1.rds.amazonaws.com -u admin -p



Option 2:

mysql -h database-1.c4xii6s2qytn.us-east-1.rds.amazonaws.com -u admin --password=***************** wordpress

```
[ec2-user@ip-10-0-0-232 wordpress]$ mysql -h database-1.c4xii6s2qytn.us-east-1.rds.amazonaws.com -u admin --password=[redacted] wordpress
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 33
Server version: 8.0.42 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [wordpress]>
```

i-026fec5852e62358b (EpicReadsWebserver)

PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

18. Configure WordPress

    Add database details

    Navigate to WordPress secret key generator available at https://api.wordpress.org/secret-key/1.1/salt/ and copy the content generated on page.

    Paste into your config file.

    Add this line at the end "define('W3TC_CONFIG_DATABASE', true);"

    sudo vi wp-config.php

```
 */

// ** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/** Database username */
define( 'DB_USER', 'admin' );

/** Database password */
define( 'DB_PASSWORD', '[redacted]' );

/** Database hostname */
define( 'DB_HOST', 'database-1.c4xii6s2qytn.us-east-1.rds.amazonaws.com' );

/** Database charset to use in creating database tables. */
define( 'DB_CHARSET', 'utf8' );

/** The database collate type. Don't change this if in doubt. */
define( 'DB_COLLATE', '' );

/**#@+
 * Authentication unique keys and salts.
 *
 * Change these to different unique phrases! You can generate these using
 * the {@link https://api.wordpress.org/secret-key/1.1/salt/ WordPress.org secret-key service}.
 *
 * You can change these at any point in time to invalidate all existing cookies.
 * This will force all users to have to log in again.
 *
 * @since 2.6.0
 */
define('AUTH_KEY',         '&`#-O4yW{9Jo.8]x_l|>|:#(LjaVw<e)4PKHyb1c`WVcq?iR7SX]CO2M%Q|i5H5G');
define('SECURE_AUTH_KEY',  'o/J79.1733F1~-~;7@Z)iD2X&|8*!LIstQ|N uGXXfe+.18D);|?eM.TFl*RJjw{');
define('LOGGED_IN_KEY',    ')-|V0Kk]k-XY[A5DLk-@%=$vJi*wX#i@,5%+-X<`U7.(qjsm).G667Z9%I+&-Qn0');
define('NONCE_KEY',        '!p=-ztf.%x2g{9?]i-={Rh~5q2*;.l&z->u`6P5UeQ=F0#Te` eh`:A$F+ShEEn(');
define('AUTH_SALT',        '_$Yh4(n ]U;nTK*JZfeO8$U h_eLIG8PE1~rZ%@ +>`#Y,b;z>`VU}~z|Z_hrgj;');
define('SECURE_AUTH_SALT', 'e9tPl|N[H{|>EtQcd2B;;.=aYL*Oi?6Pu[ML%A5,nsD/O,LzB0ZH%|#E8alPKzF/');
define('LOGGED_IN_SALT',   '3(3<1nCMlxw.KhTbjd~RqV ~I *#dB!<)ehc8JF<p|)On&j c/pO<HVU+|B6t9;j');
define('NONCE_SALT',       'cn2F,oWj<(q/.!+_-m58<%90x;n[$bgYQo-o-{PkZ|8+:7&&lNr?D-BlBV&{l~R3');
/**#@-*/

/**
```

19. Commands to deploy WordPress.

    sudo yum install php-xml -y

    cd ..

    sudo cp -r wordpress/* /var/www/html

    sudo chown -R apache:apache /var/www/html

    sudo systemctl start httpd.service

sudo systemctl restart php-fpm

```
[ec2-user@ip-10-0-0-232 wordpress]$ sudo yum install php-xml -y
Last metadata expiration check: 1:39:12 ago on Sun Sep 21 06:35:43 2025.
Package php8.4-xml-8.4.10-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-232 wordpress]$ cd ..
[ec2-user@ip-10-0-0-232 html]$ sudo cp -r wordpress/* /var/www/html
[ec2-user@ip-10-0-0-232 html]$ sudo chown -R apache:apache /var/www/html
[ec2-user@ip-10-0-0-232 html]$ sudo systemctl start httpd.service
[ec2-user@ip-10-0-0-232 html]$ sudo systemctl restart php-fpm
[ec2-user@ip-10-0-0-232 html]$
```

i-026fec5852e62358b (EpicReadsWebserver)

PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

Restart HTTPD if you don’t see the correct WordPress setup from next step.

sudo systemctl restart httpd

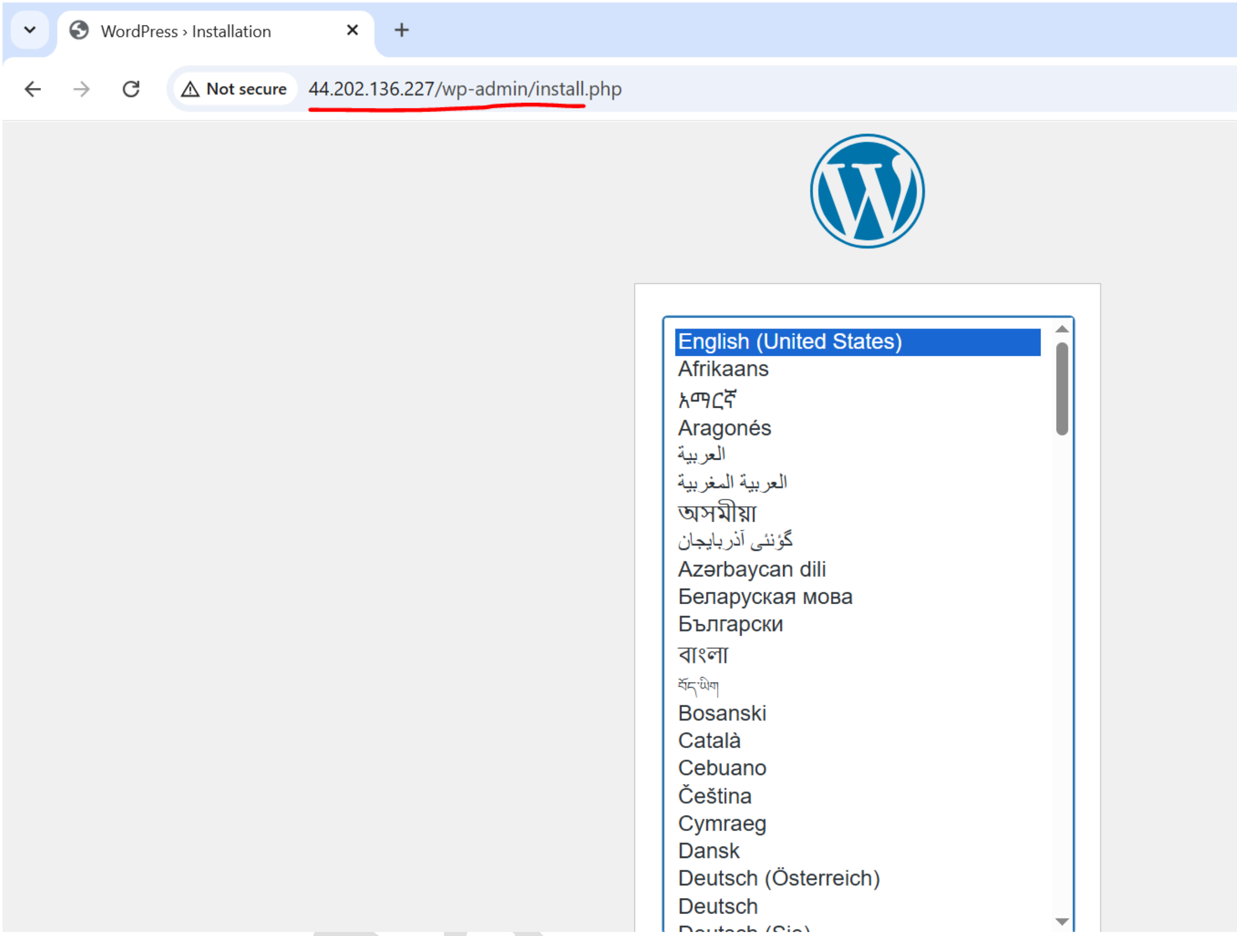
sudo systemctl status httpd

```
[ec2-user@ip-10-0-0-232 html]$ sudo systemctl restart httpd
[ec2-user@ip-10-0-0-232 html]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Sun 2025-09-21 08:33:17 UTC; 21s ago
       Docs: man:httpd.service(8)
   Main PID: 34510 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1111)
     Memory: 13.0M
        CPU: 76ms
     CGroup: /system.slice/httpd.service
             ├─34510 /usr/sbin/httpd -DFOREGROUND
             ├─34511 /usr/sbin/httpd -DFOREGROUND
             ├─34512 /usr/sbin/httpd -DFOREGROUND
             ├─34513 /usr/sbin/httpd -DFOREGROUND
             └─34514 /usr/sbin/httpd -DFOREGROUND

Sep 21 08:33:17 ip-10-0-0-232.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 21 08:33:17 ip-10-0-0-232.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 21 08:33:17 ip-10-0-0-232.ec2.internal httpd[34510]: Server configured, listening on: port 80
[ec2-user@ip-10-0-0-232 html]$
```

i-026fec5852e62358b (EpicReadsWebserver)

PublicIPs: 44.202.136.227 PrivateIPs: 10.0.0.232

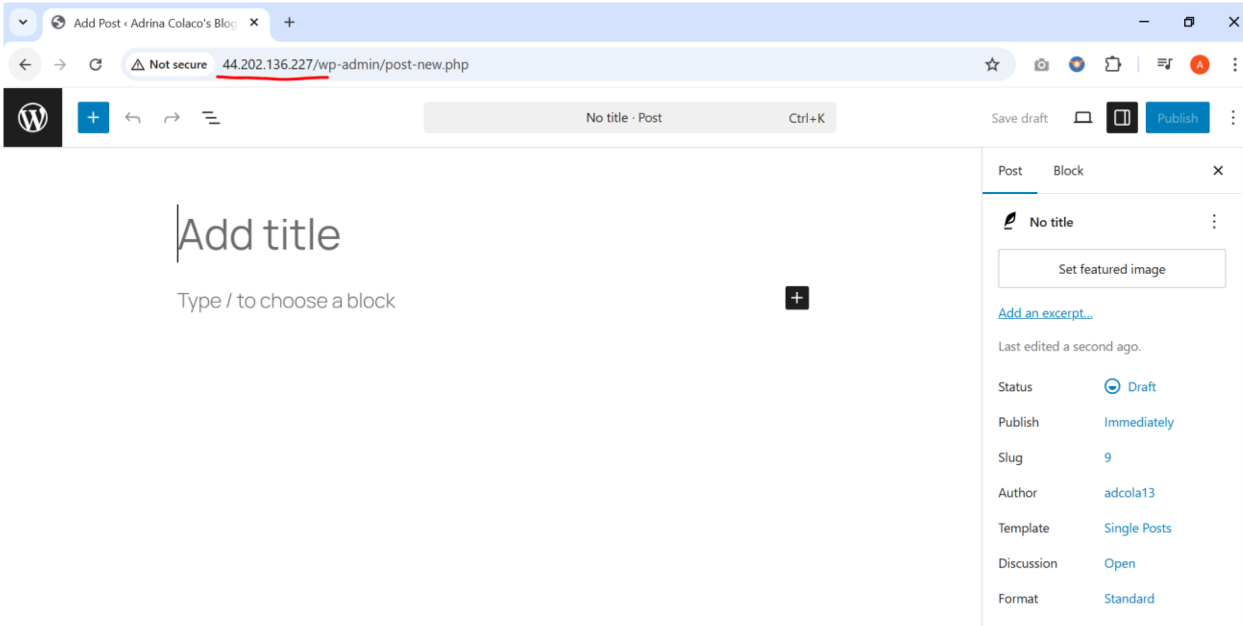
20. Verify and complete WordPress Setup on public IP of EC2 webserver
    http://<PublicIP>/wp-admin/



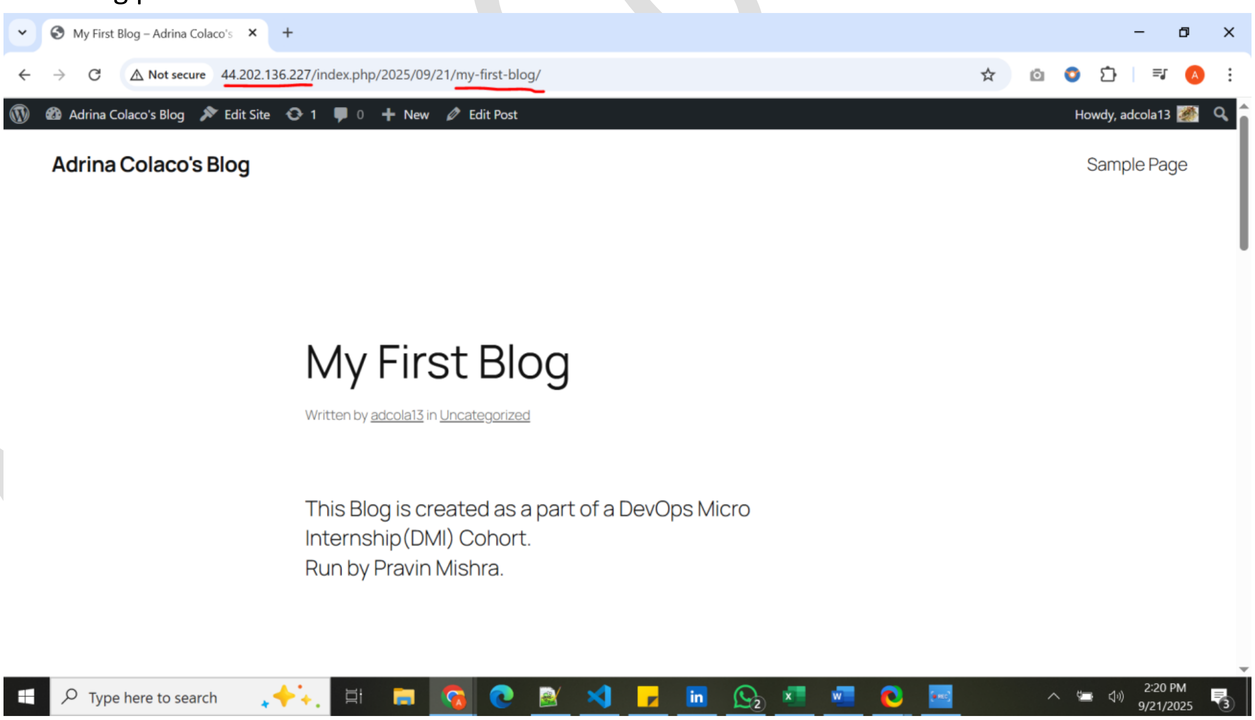
Complete the WordPress setup as shown in the video below.

TwoTier Application.mp4

21. Create a blog post.

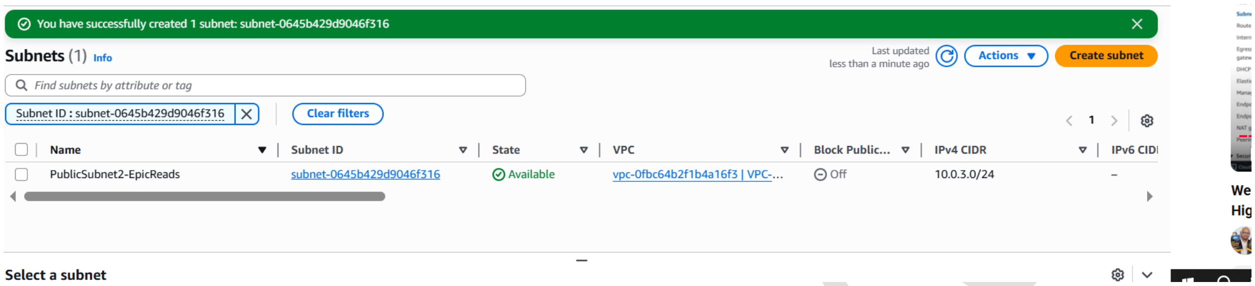


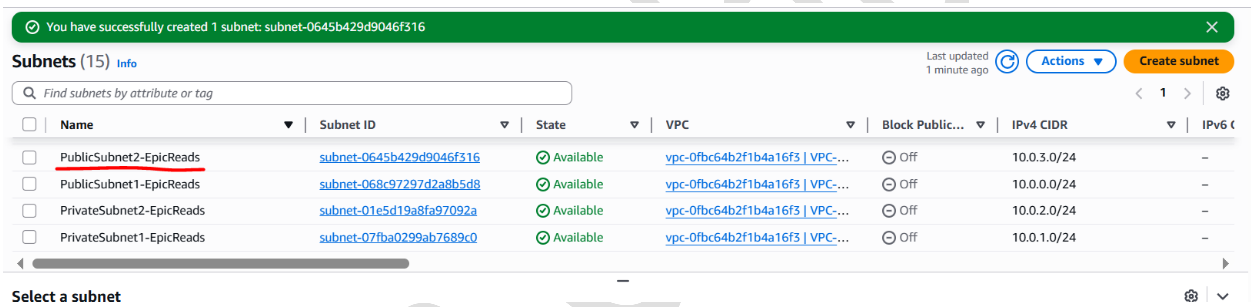
Final blog post should look like:



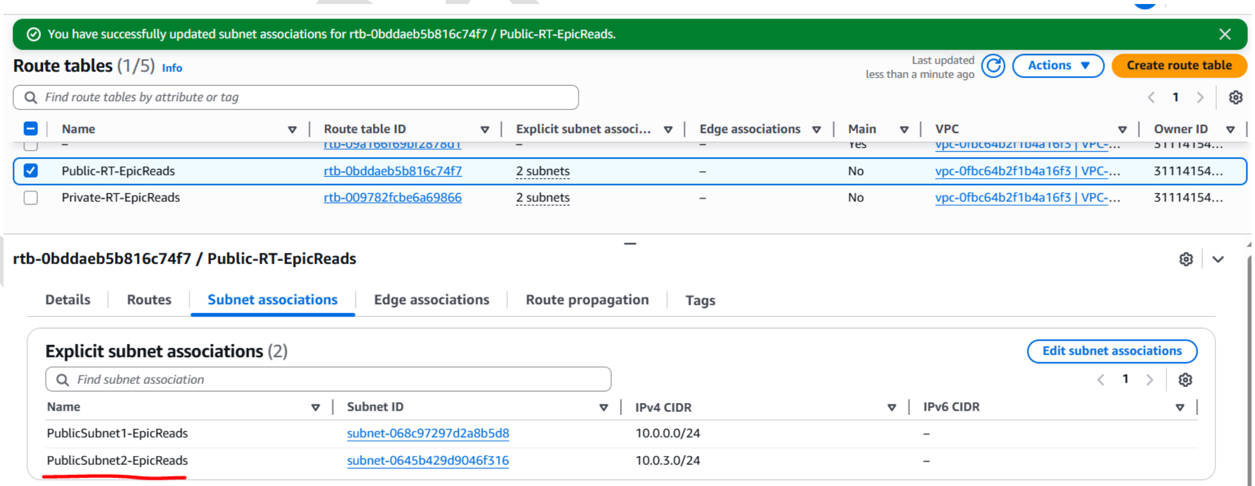
Assignment 18 – Making Two-Tier Application HA (Part 1, Part 2, Part 3)

1. Create another Public Subnet

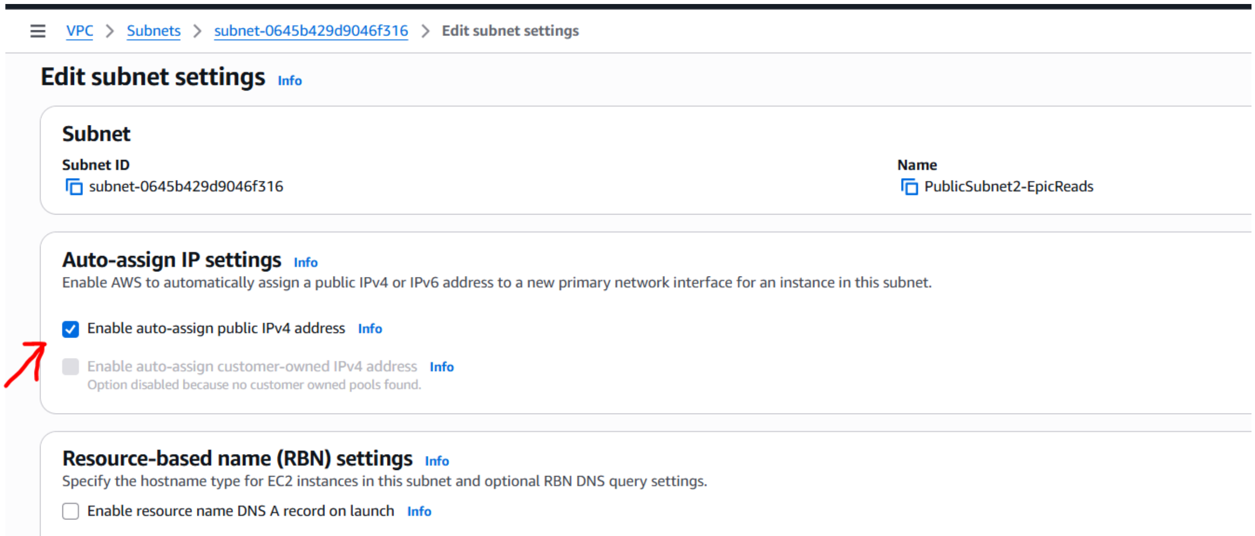




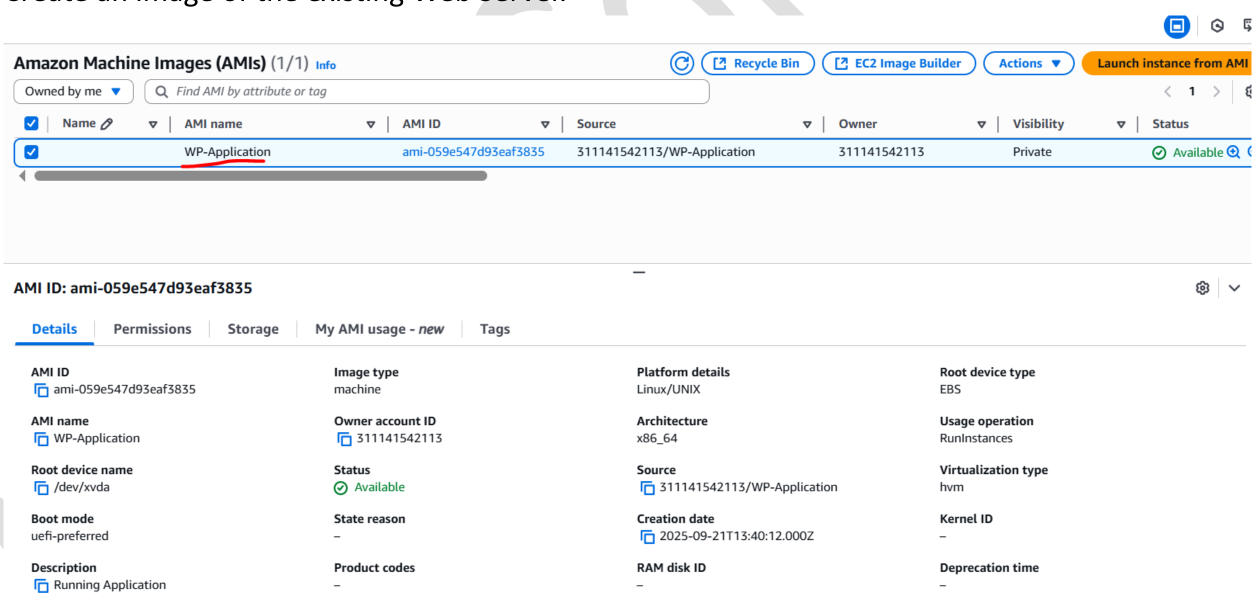
2. Add the newly created Public Subnet to Public Route Table.



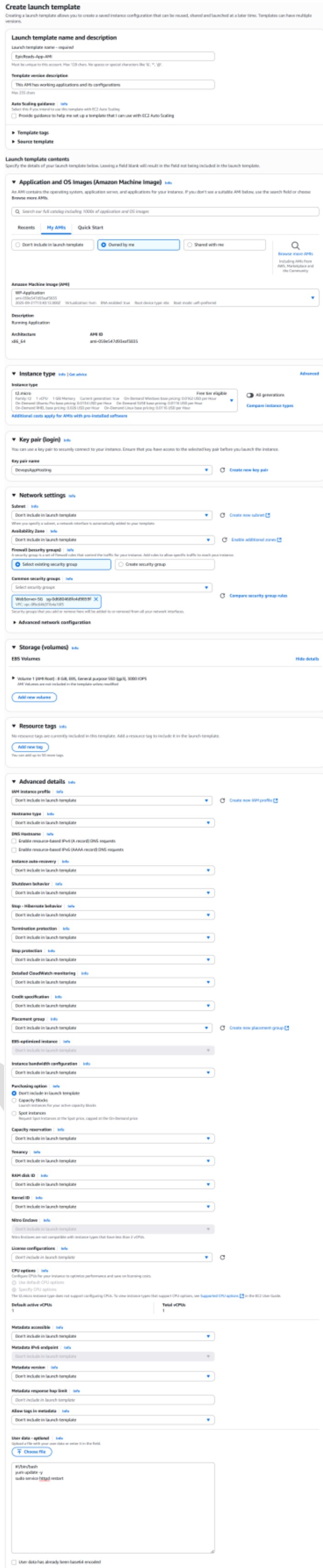
3. Modify setting for both Public subnets to enable "auto assign" IP address.

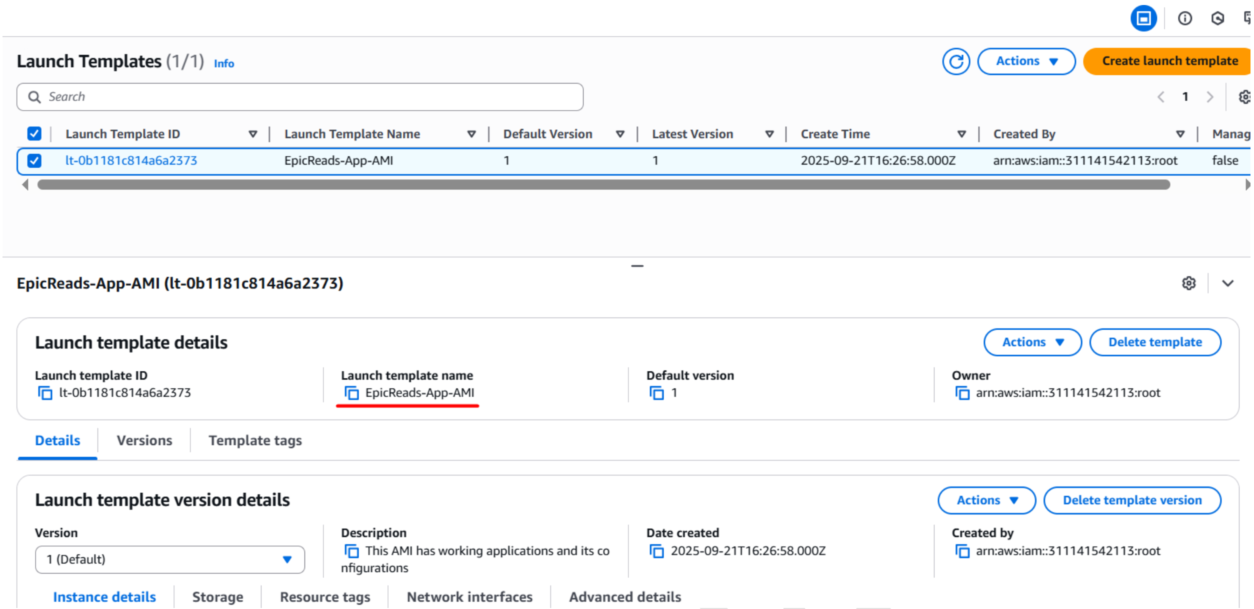


4. Create an image of the existing Web Server.

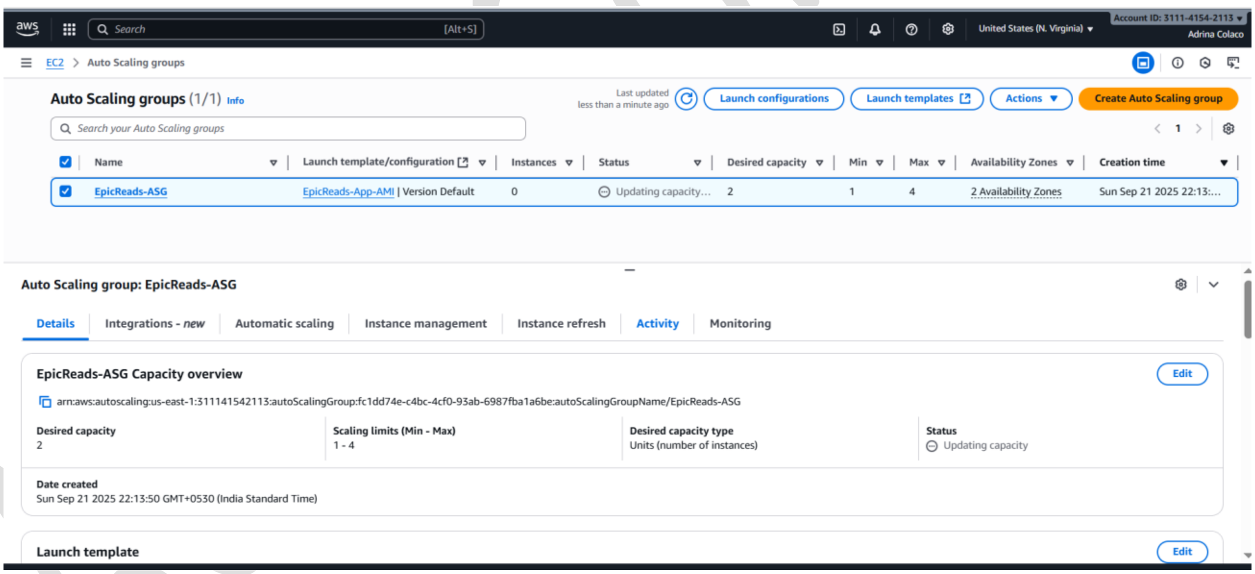


5. Launch Template

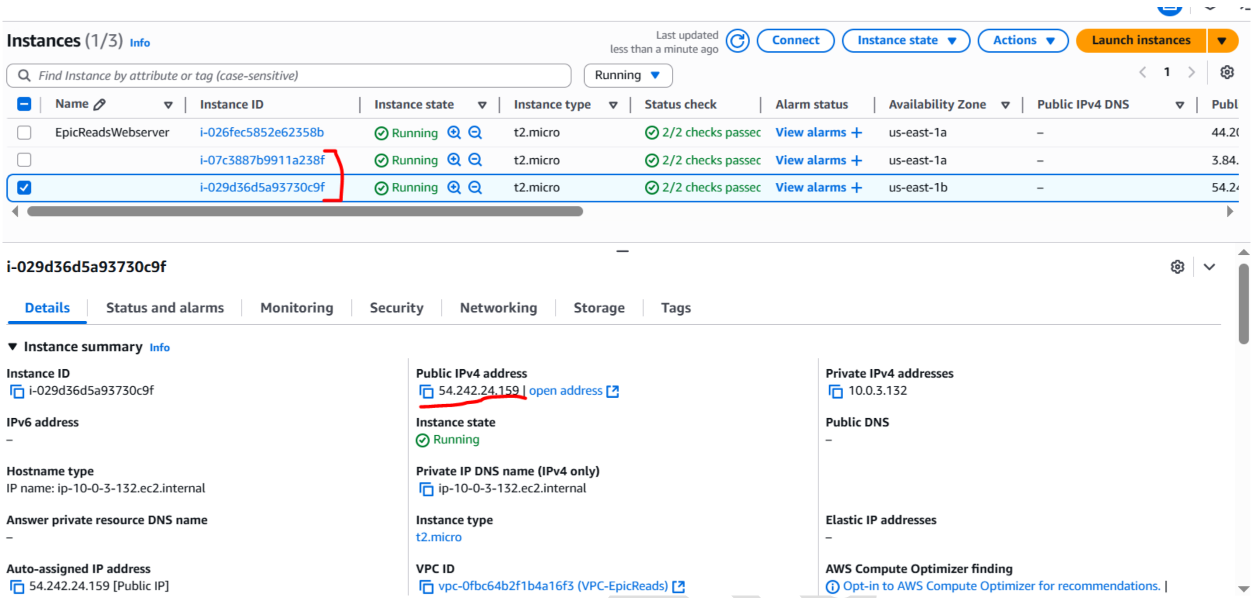

Create launch template
Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.
Launch template name and description
Launch template name - required
EpicReads-App-AMI
Template version description
This AMI has working applications and its configurations
Auto Scaling guidance
Provide guidance to help me set up a template that I can use with EC2 Auto Scaling
Template tags
Source template
Launch template contents
Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.
Application and OS Images (Amazon Machine Image)
Recents
My AMIs
Quick Start
Don't include in launch template
Owned by me
Shared with me
Browse more AMIs
Amazon Machine Image (AMI)
WP-Application
Description
Running Application
Architecture
x86_64
AMI ID
Instance type
t2.micro
Free tier eligible
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
Network settings
Storage (volumes)
Resource tags
Advanced details
Metadata accessible
Metadata IPv6 endpoint
Metadata version
Metadata response hop limit
Allow tags in metadata
User data - optional
Choose file
#!/bin/bash
yum update -y
sudo service httpd restart
User data has already been base64 encoded



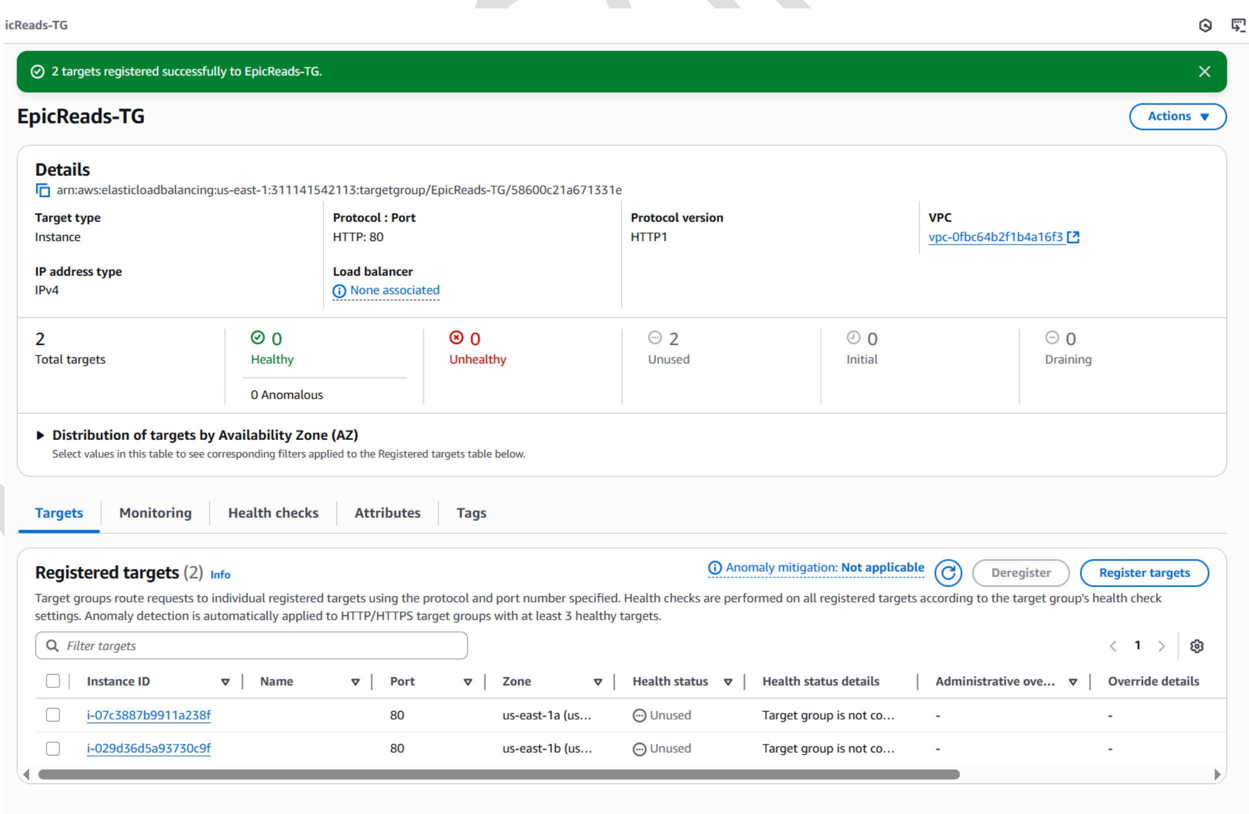
6. Create Auto Scaling Group



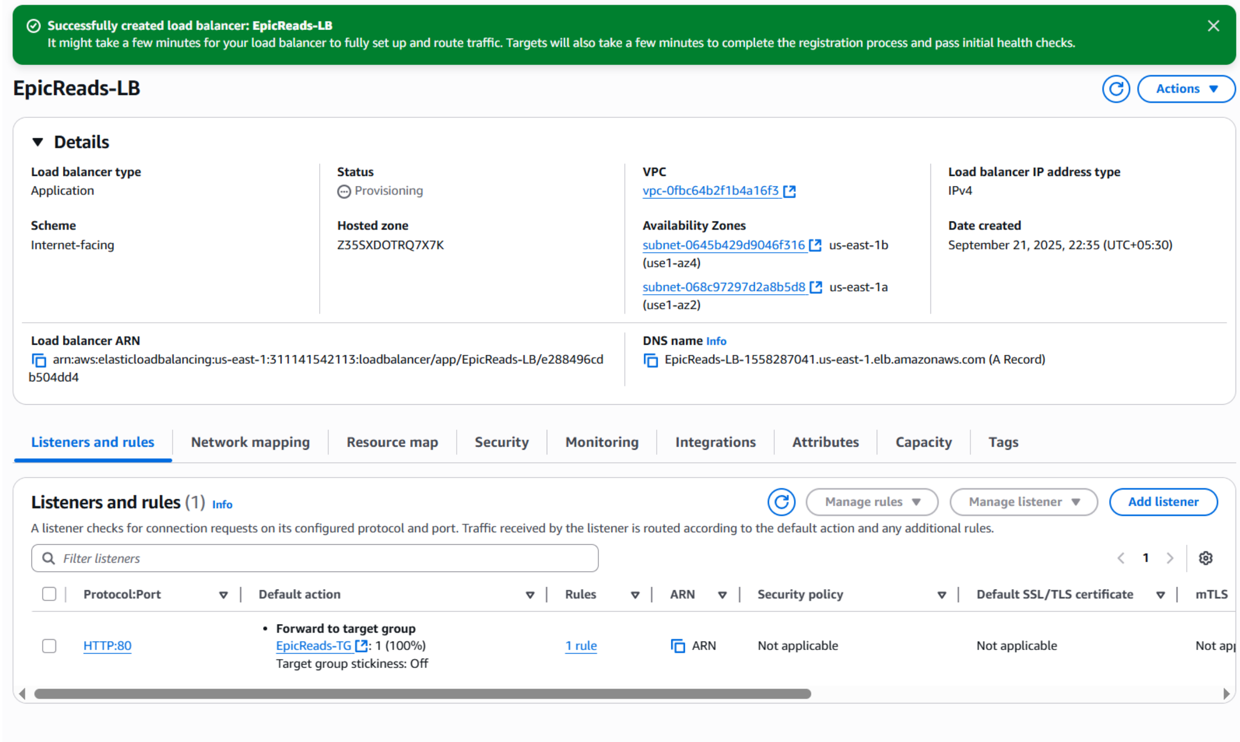
7. Verify the new instances that are created as a part of autoscaling



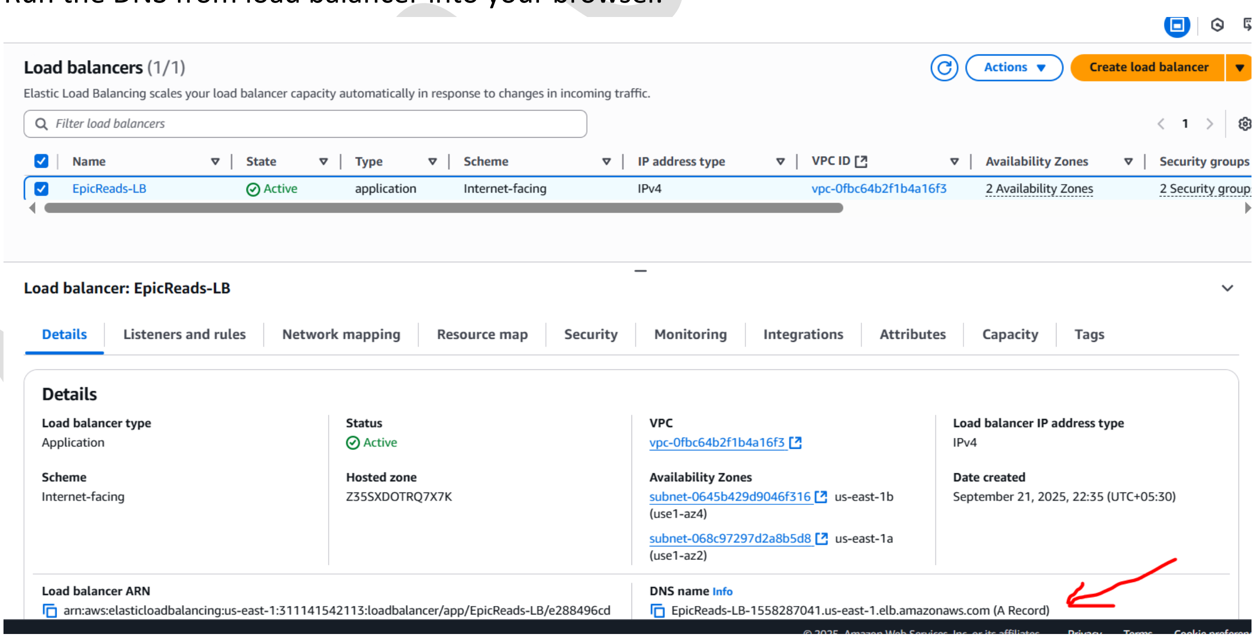
8. Create Target Group



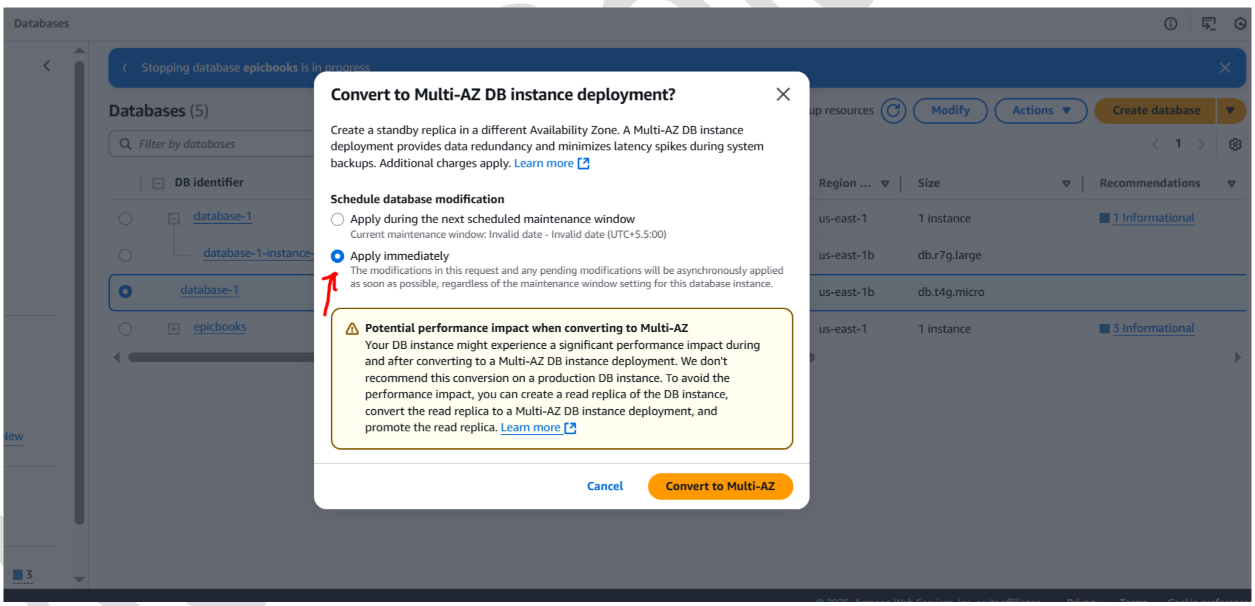
9. Launch Load balancer



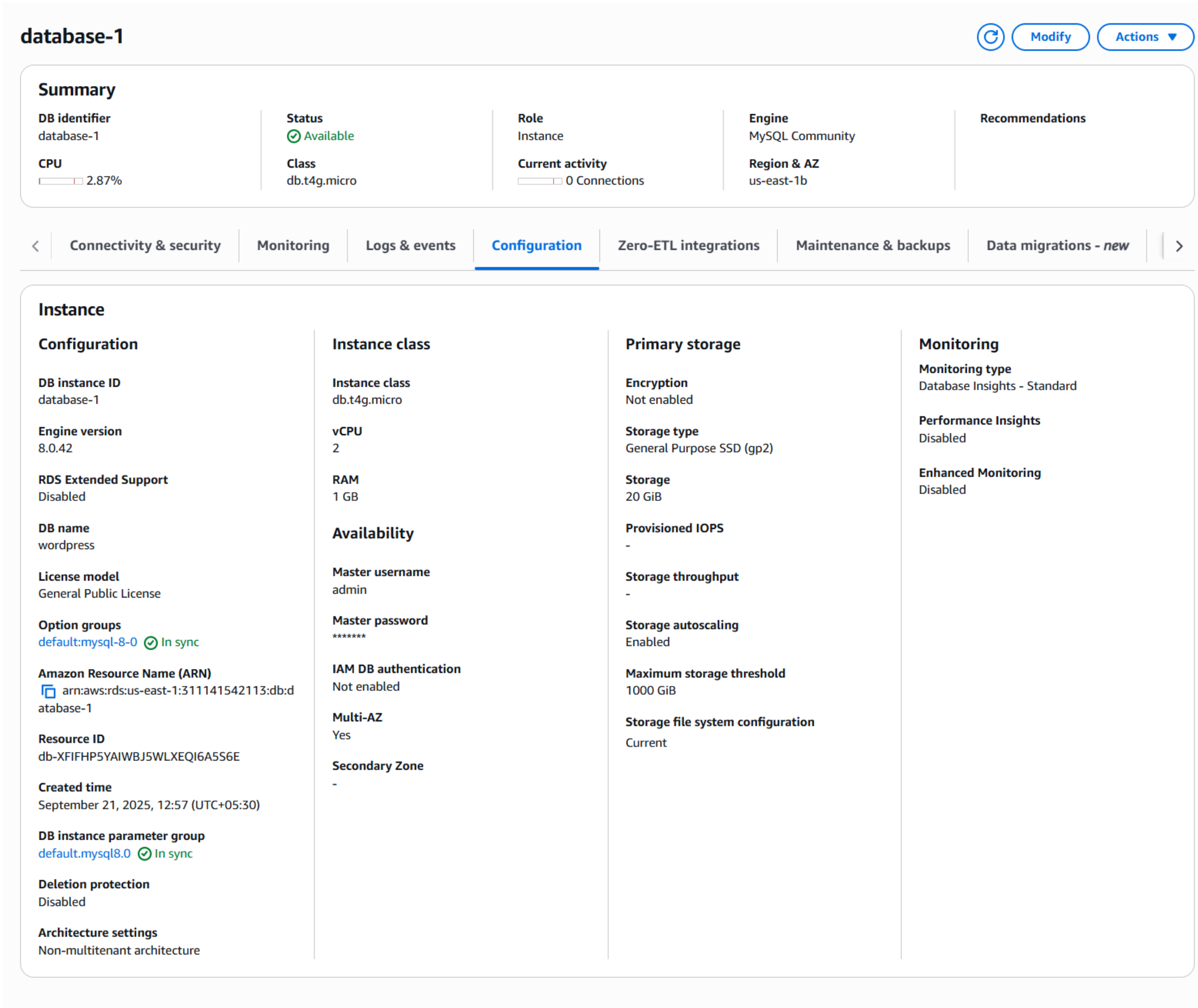
10. Run the DNS from load balancer into your browser.



11. Make the database highly available by "Convert to Multi AZ Deployment".

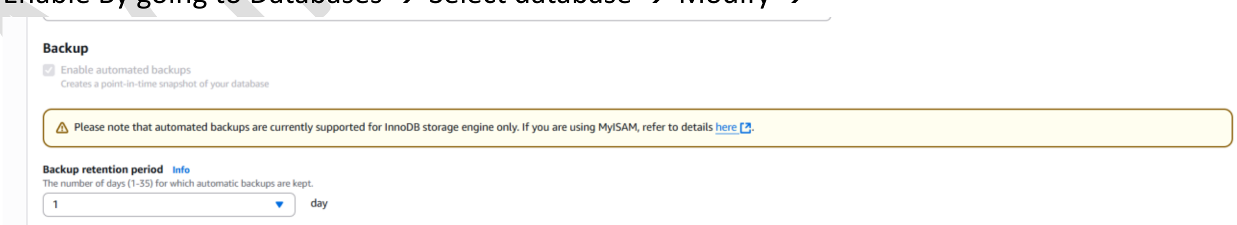


Look out for Availability and Multi-AZ status.



12. Create a Read replica of the database.

Enable By going to Databases → Select database → Modify →



Actions → Read Replica

13. Run the DNS from load balancer into your browser again.

LinkedIn Posts:

https://lnkd.in/p/g9vGJk5P

https://lnkd.in/p/gb3vNEyq

https://lnkd.in/p/gkPksCrm

https://lnkd.in/p/gi_24RxV

Blogs:

https://medium.com/@adcola13/how-to-host-a-static-website-using-amazon-s3-a-complete-setup-guide-e7a7ee3d82cd

https://medium.com/@adcola13/deploying-a-basic-two-tier-application-architecture-for-a-wordpress-blog-f68b5b3d8f79

The End

P.S. This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by Pravin Mishra. You can start your DevOps journey for free from his YouTube Playlist.